



PRODUCT DOCUMENTATION

Forge MES Product Cookbook

A visual, self-serve guide that explains manufacturing in plain language and walks from factory setup to a traceable finished unit.

48

guided articles

10

ordered setup stages

Updated 24 July 2026 | Searchable PDF | Clickable contents

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GUIDED STAGE

Start Here

Learn the manufacturing basics, understand how Forge records connect, and see the complete journey before configuring anything.

<https://www.fynd.com/cookbooks/forge/start-here>

Follow this stage in order

- 1 [Forge MES in Plain Language](#) - Understand what Forge MES does without needing a manufacturing background.
- 2 [Manufacturing Basics for New Users](#) - Learn the small set of factory concepts used throughout the cookbook.
- 3 [How Forge Data Connects](#) - See the dependency chain that connects factory setup to executable production.
- 4 [Complete Forge Setup Journey](#) - Follow the recommended order for configuring Forge and releasing the first production run.
- 5 [How to Use This Cookbook](#) - Use the cookbook as a guided setup path, task reference, and troubleshooting tool.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Forge MES in Plain Language

Understand what Forge MES does without needing a manufacturing background.

<https://www.fynd.com/cookbooks/forge/start-here/forge-mes-in-plain-language>

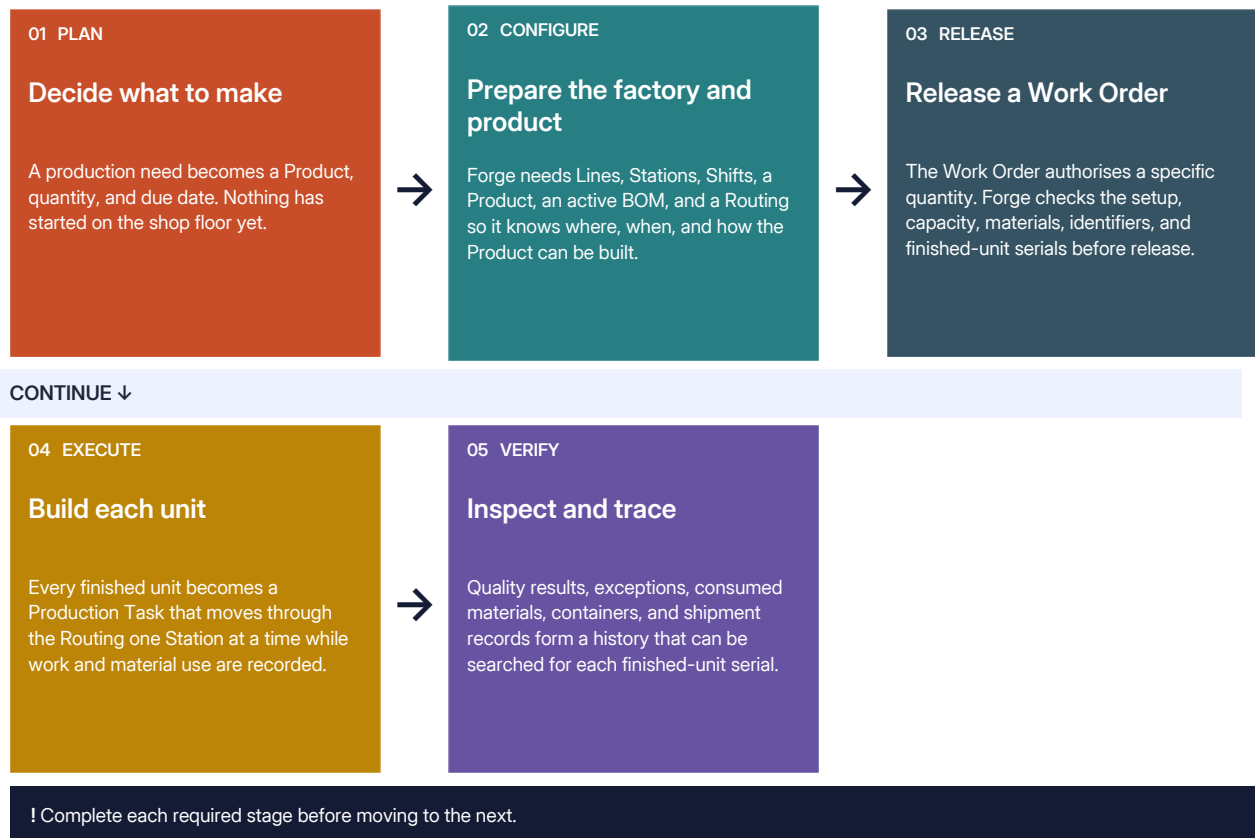
Requirement	Required
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Before you begin

- No earlier cookbook topic is required.

How it works

Forge is the digital control layer between a production plan and the real work performed in the factory. This is the full journey of one production run.



Why this connection matters: Forge MES manages manufacturing execution; it does not replace the physical factory.

What to do

- 1 Use Forge to turn a production plan into controlled work on real factory lines.
- 2 Give every unit a route, a production task, and a history that can be traced later.
- 3 Use checks, repair paths, packaging, and dashboards to keep production reliable.

Rules to remember

- Forge MES manages manufacturing execution; it does not replace the physical factory.
- Master data describes what can happen. Transaction data records what actually happened.
- The setup sequence matters because downstream records depend on upstream records.

Ready when

- ☐ I can explain the difference between setup data and production activity.
- ☐ I understand that a finished unit moves through a routing one station at a time.
- ☐ I know that Forge keeps a traceable record for every released unit.

Manufacturing Basics for New Users

Learn the small set of factory concepts used throughout the cookbook.

<https://www.fynd.com/cookbooks/forge/start-here/manufacturing-basics>

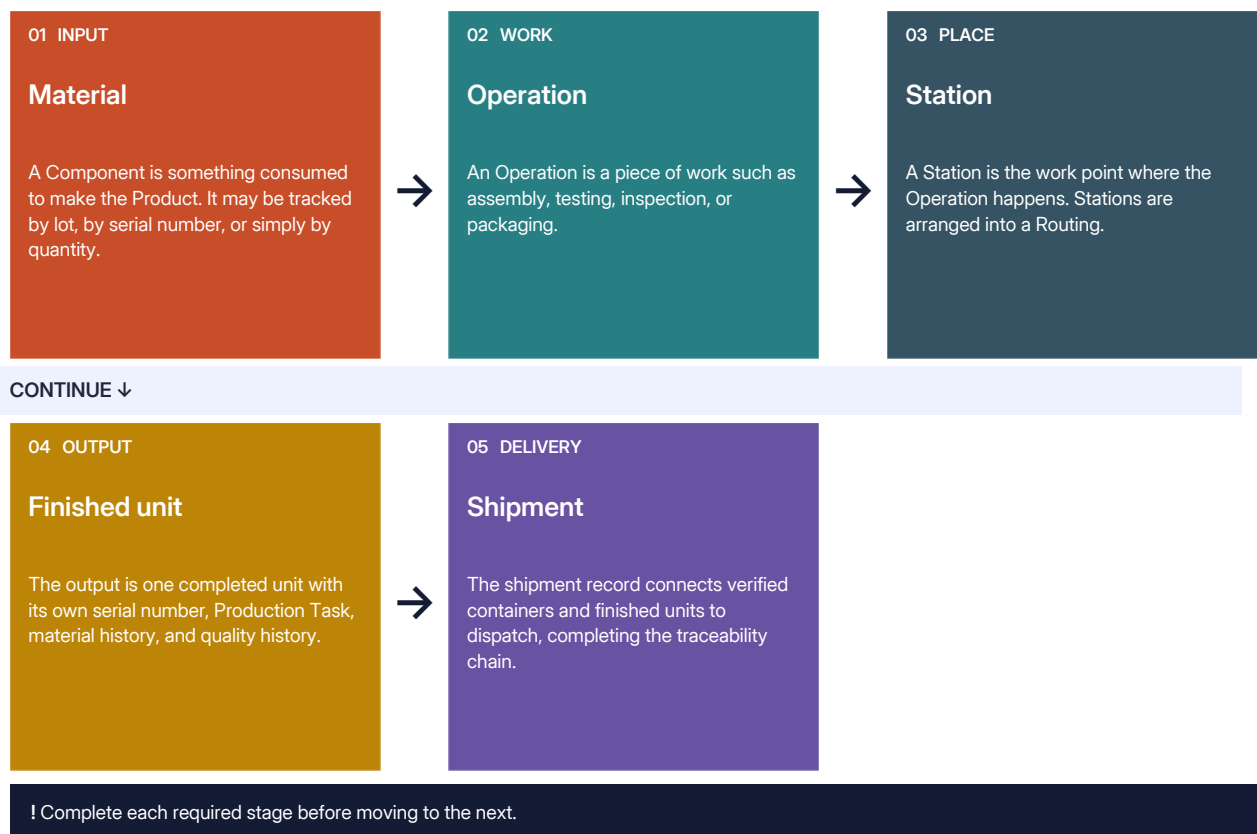
Requirement	Required
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Before you begin

- No earlier cookbook topic is required.

How it works

Imagine one physical product moving from raw material to a customer. These concepts describe what happens to it.



Why this connection matters: A Line is a production lane; a Station is a specific place where an operation happens.

What to do

- 1 A product is the item the factory intends to make.
- 2 A bill of materials lists the components required to make that product.
- 3 A routing lists the operations and stations the unit must pass through.
- 4 A work order authorises a quantity of that product to be produced.

- 5 A production task represents one finished unit moving through the route.

Rules to remember

- A Line is a production lane; a Station is a specific place where an operation happens.
- A component can be scanned by serial number or consumed by quantity.
- A failed operation can send the unit to repair, rework, hold, or scrap depending on configuration.

Ready when

- ☐ I can distinguish a product, component, BOM, routing, work order, and production task.
- ☐ I understand the difference between a Line and a Station.
- ☐ I know why traceability begins before production is released.

How Forge Data Connects

See the dependency chain that connects factory setup to executable production.

<https://www.fynd.com/cookbooks/forge/start-here/how-forge-data-connects>

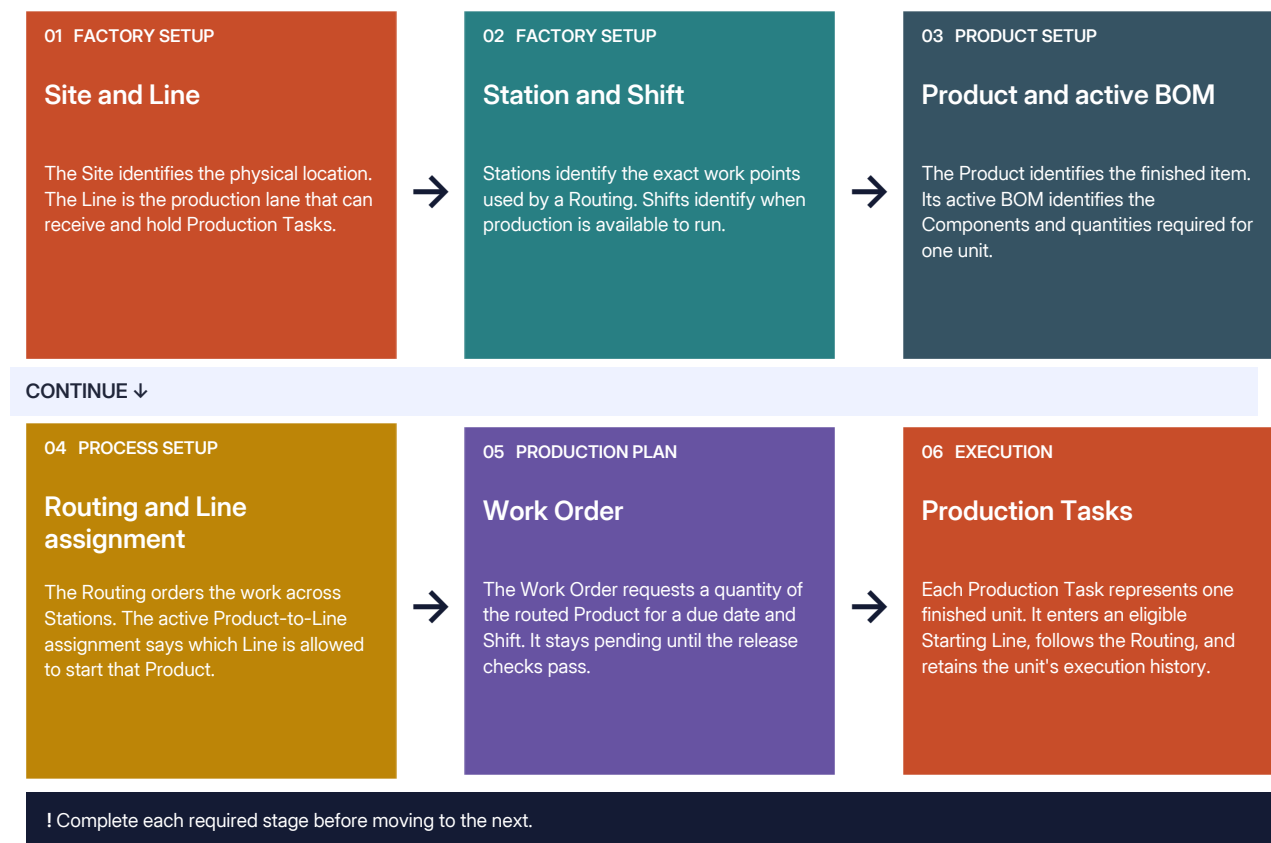
Requirement	Required
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Before you begin

- Forge MES in Plain Language
- Manufacturing Basics for New Users

How it works

Forge uses two setup branches—where work happens and what is being made. They meet in the Routing and Line assignment before a Work Order can run.



Why this connection matters: A missing prerequisite blocks the next record even when the next screen is visible.

What to do

- 1 Sites locate Stations and can also group Lines.
- 2 Products use BOMs to define material requirements.
- 3 Routings connect operations to Stations and require an active BOM.

- 4 Product-to-Line assignments identify where a product can start production.
- 5 Releasing a Work Order creates unit-level Production Tasks on eligible Lines.

Rules to remember

- A missing prerequisite blocks the next record even when the next screen is visible.
- Projects and Product Families are recommended organisational layers, not hard execution blockers.
- Production Orders are optional grouping records; Work Orders create executable production.

Ready when

- [] I can follow the dependency chain from Site to Production Task.
- [] I know which organisational records are recommended.
- [] I understand why a Production Order alone does not start production.

Complete Forge Setup Journey

Follow the recommended order for configuring Forge and releasing the first production run.

<https://www.fynd.com/cookbooks/forge/start-here/complete-setup-journey>

Requirement	Required
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Before you begin

- How Forge Data Connects

How it works

Use this as the setup map for a new Forge environment. Do not skip ahead: every required stage supplies information used by the next stage.



! Complete each required stage before moving to the next.

Why this connection matters: Do not create Work Orders until the Product has both a Routing and an active Line assignment.

What to do

- 1 Review the Plant and create the required Sites.
- 2 Create Lines, mark the correct initial Line as a Starting Line, and configure capacity.
- 3 Create operation Stations, dedicated Repair Stations, and Shifts.
- 4 Create recommended Projects and Product Families, followed by Components and Products.
- 5 Review the draft BOM created with the Product and activate it.
- 6 Build and activate a Routing, then assign the Product to eligible Lines.
- 7 Create a Work Order with at least one Shift, generate unit serials, validate readiness, and release.

Rules to remember

- Do not create Work Orders until the Product has both a Routing and an active Line assignment.
- Do not release production until the initial Line is marked as a Starting Line.
- Do not bypass readiness warnings without understanding their effect on later Stations.

Ready when

- ☐ Every setup stage has an owner in my implementation plan.
- ☐ Dependencies are complete before the first Work Order is created.
- ☐ A small trial Work Order will be used before full-volume production.

How to Use This Cookbook

Use the cookbook as a guided setup path, task reference, and troubleshooting tool.

<https://www.fynd.com/cookbooks/forge/start-here/how-to-use-this-cookbook>

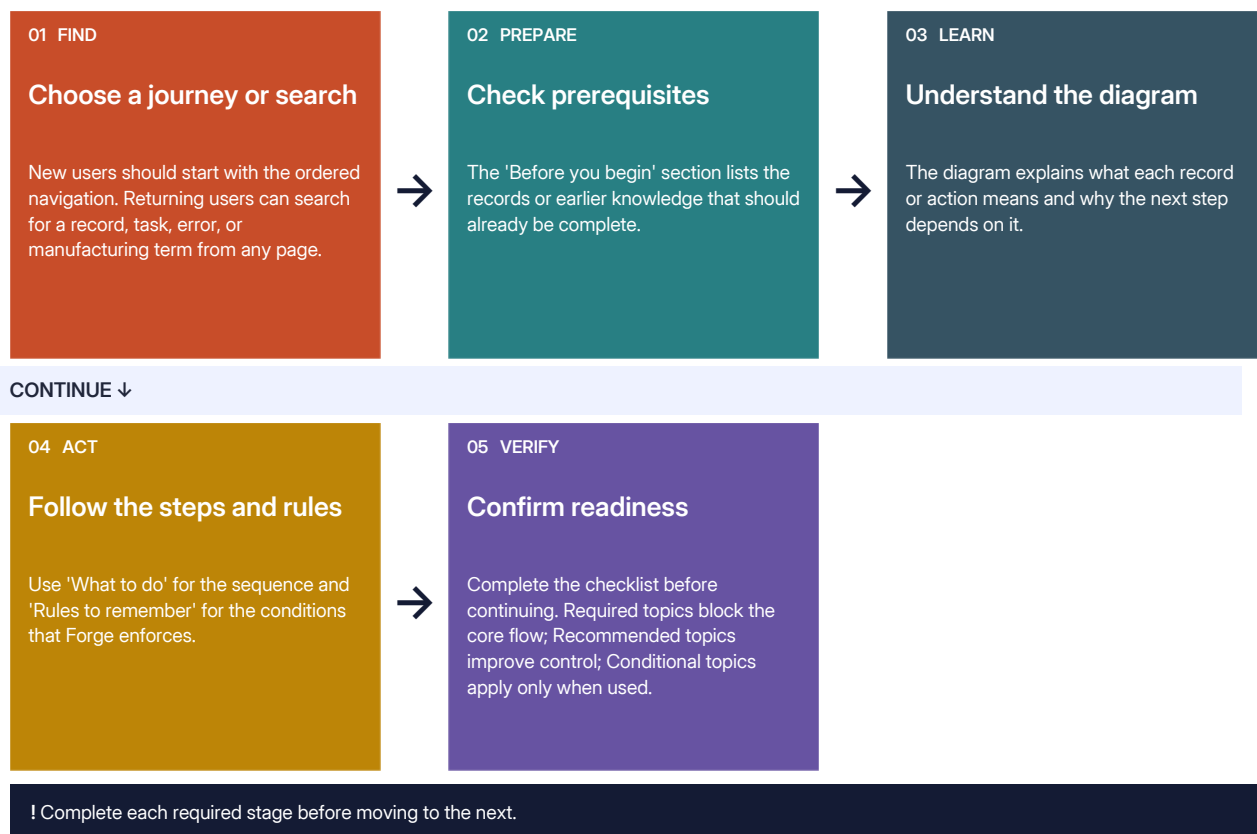
Requirement	Recommended
-------------	-------------

Before you begin

- No earlier cookbook topic is required.

How it works

You can read the cookbook from beginning to end or enter through global search. Every guide uses the same five-part learning pattern.



Why this connection matters: Required means Forge blocks or cannot reliably complete the downstream action without it.

What to do

- 1 New users should follow the sections in the order shown in the navigation.
- 2 Experienced users can use global search from any page to find a record or task.
- 3 Use the Required, Recommended, and Conditional labels to understand priority.

- 4 Use the readiness checklist at the end of each guide before continuing.

Rules to remember

- Required means Forge blocks or cannot reliably complete the downstream action without it.
- Recommended means Forge may allow the action, but the setup improves organisation and control.
- Conditional means the setup is needed only when that manufacturing capability is used.

Ready when

- ☐ I know how to search from every cookbook page.
- ☐ I understand the three requirement labels.
- ☐ I know where to download the complete cookbook as a PDF.

GUIDED STAGE

Configure Your Factory

Create the physical structure Forge needs: Sites, Lines, Stations, Repair Stations, capacity, and Shifts.

<https://www.fynd.com/cookbooks/forge/configure-factory>

Follow this stage in order

- 1 [The Forge Factory Model](#) - Understand how Plant, Site, Line, and Station represent a real manufacturing unit.
- 2 [Create Sites](#) - Create the physical locations required before Stations can be configured.
- 3 [Create Production Lines](#) - Create the production lanes that will receive and balance unit-level Production Tasks.
- 4 [Create Stations and Repair Stations](#) - Create operational work points and the separate repair destinations required by Routings.
- 5 [Create Shifts](#) - Define when production runs and make Shifts available for Work Order assignment.
- 6 [Factory Readiness Checklist](#) - Confirm the physical factory model is complete before creating production master data.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

The Forge Factory Model

Understand how Plant, Site, Line, and Station represent a real manufacturing unit.

<https://www.fynd.com/cookbooks/forge/configure-factory/factory-model>

Requirement	Required
-------------	----------

Before you begin

- Complete Forge Setup Journey

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Stations require a Site.

What to do

- 1 The Plant is the overall manufacturing context used to isolate records.
- 2 A Site represents a physical location within the Plant.
- 3 A Line represents a production lane and receives Production Tasks.
- 4 A Station represents a specific operational point used in a Routing.

Rules to remember

- Stations require a Site.
- Lines can reference a Site, but the current Line setup does not require it.
- A Station can exist without a Line assignment, although linking the physical structure is recommended.

Ready when

- [] Every operation has a clear physical Site.
- [] Production lanes are represented as Lines.
- [] Work areas are represented as Stations rather than additional Lines.

Create Sites

Create the physical locations required before Stations can be configured.

<https://www.fynd.com/cookbooks/forge/configure-factory/sites>

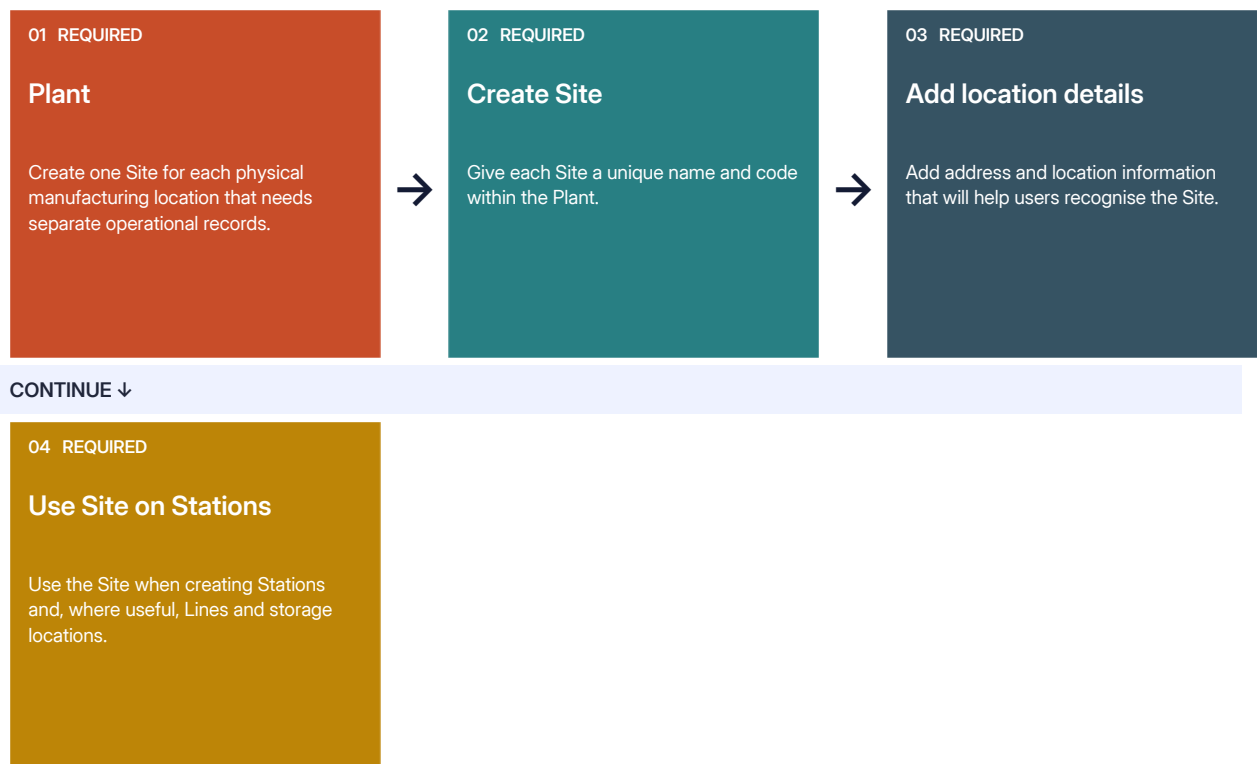
Requirement	Required
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Before you begin

- The Forge Factory Model

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



! Complete each required stage before moving to the next.

Why this connection matters: A Station cannot be created without a valid Site.

What to do

- 1 Create one Site for each physical manufacturing location that needs separate operational records.
- 2 Give each Site a unique name and code within the Plant.
- 3 Add address and location information that will help users recognise the Site.
- 4 Use the Site when creating Stations and, where useful, Lines and storage locations.

Rules to remember

- A Station cannot be created without a valid Site.
- Do not create a separate Site for every room or workstation; use Stations for that detail.
- Use globally understandable names rather than local abbreviations without explanation.

Ready when

- ☐ Site names and codes are unique.
- ☐ Every planned Station can be mapped to a Site.
- ☐ Users can distinguish Sites without relying only on codes.

Create Production Lines

Create the production lanes that will receive and balance unit-level Production Tasks.

<https://www.fynd.com/cookbooks/forgem/configure-factory/lines>

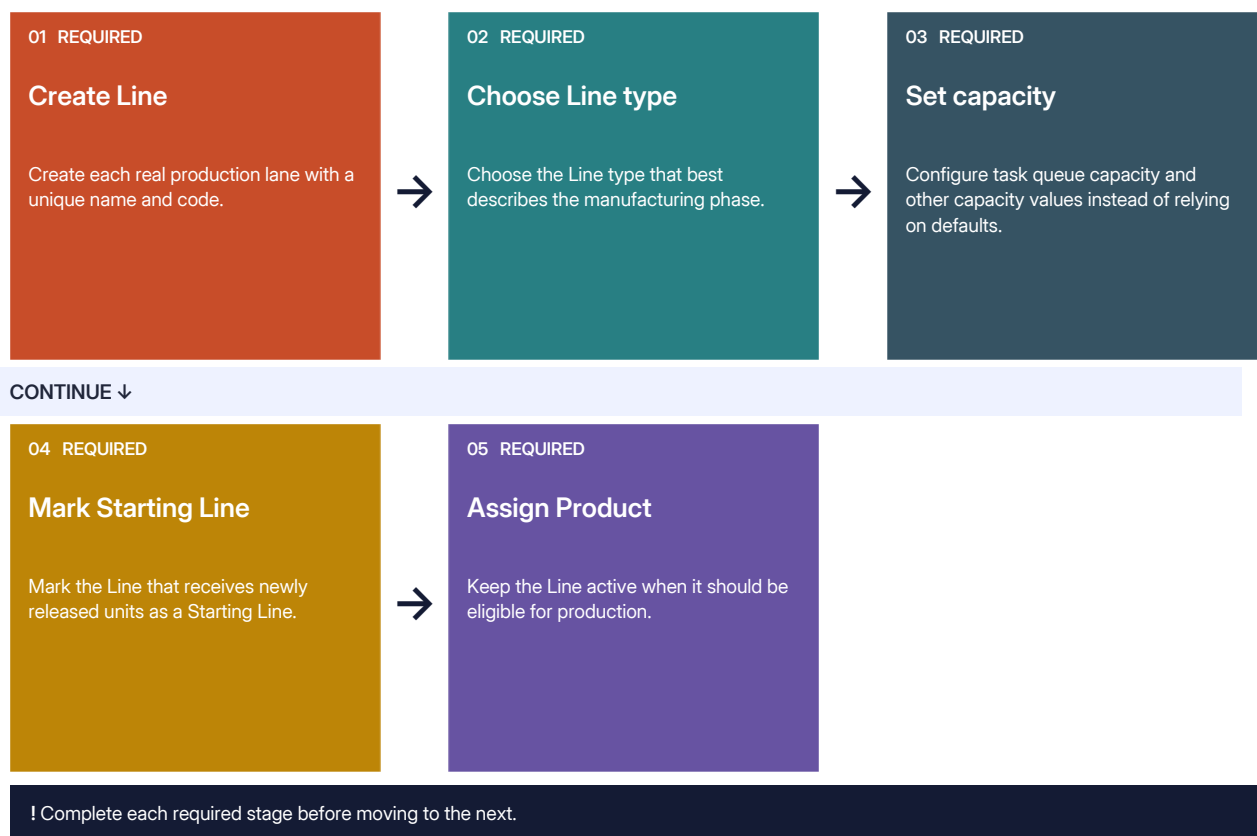
Requirement	Required
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Before you begin

- The Forge Factory Model

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Product needs at least one active Line assignment before a Work Order can be created.

What to do

- Create each real production lane with a unique name and code.
- Choose the Line type that best describes the manufacturing phase.
- Configure task queue capacity and other capacity values instead of relying on defaults.
- Mark the Line that receives newly released units as a Starting Line.

- 5 Keep the Line active when it should be eligible for production.

Rules to remember

- A Product needs at least one active Line assignment before a Work Order can be created.
- Normal initial release uses only eligible Lines marked as Starting Lines.
- Forge can fall back to a queue capacity of 100, but explicit capacity is recommended.

Ready when

- [] Every production lane has a unique Line.
- [] The initial Line is marked as a Starting Line.
- [] Capacity reflects the number of active and waiting tasks the Line can accept.

Create Stations and Repair Stations

Create operational work points and the separate repair destinations required by Routings.

<https://www.fynd.com/cookbooks/forge/configure-factory/stations-and-repair-stations>

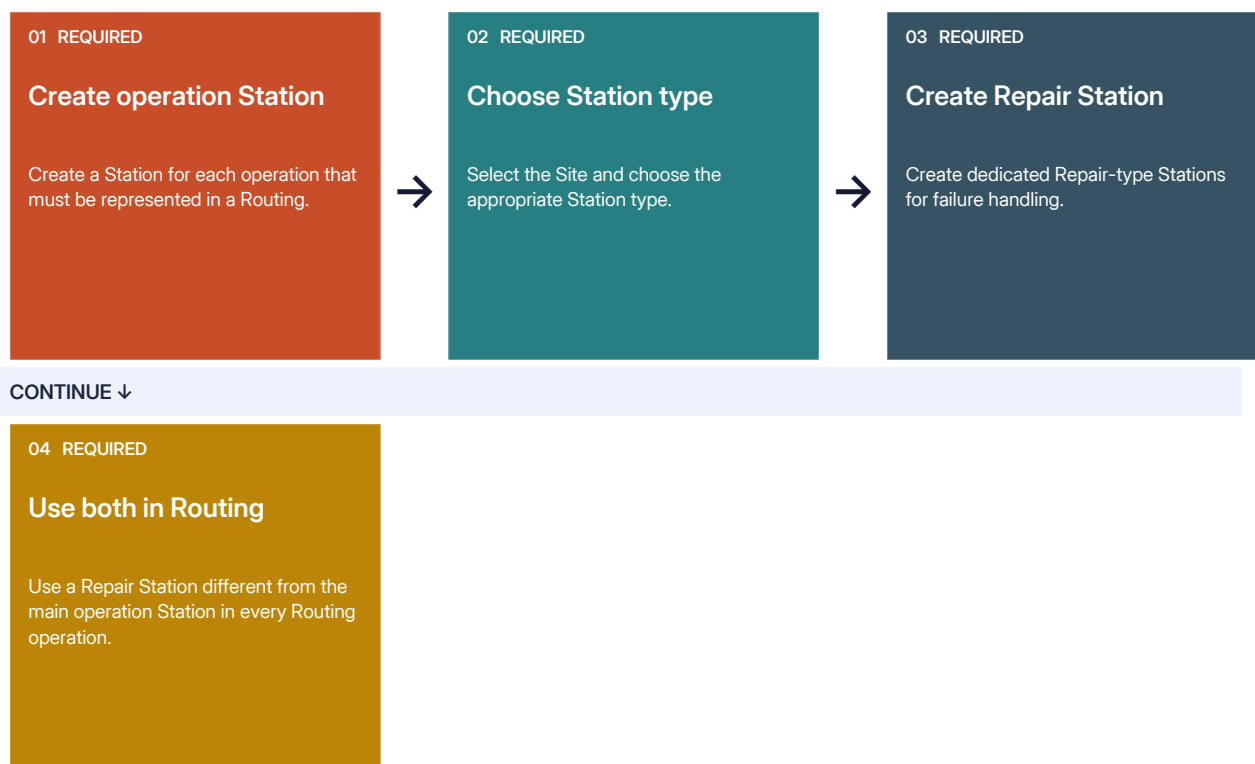
Requirement	Required
-------------	----------

Before you begin

- Create Sites

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



! Complete each required stage before moving to the next.

Why this connection matters: Station name, code, and Site are required.

What to do

- 1 Create a Station for each operation that must be represented in a Routing.
- 2 Select the Site and choose the appropriate Station type.
- 3 Create dedicated Repair-type Stations for failure handling.
- 4 Use a Repair Station different from the main operation Station in every Routing operation.

Rules to remember

- Station name, code, and Site are required.
- Every operation node in the current Routing builder requires a Repair Station.
- The Repair Station cannot be the same Station as the operation itself.
- Automatic Stations are used for configured automation; regular Stations expect operator interaction.

Ready when

- ☐ Every planned operation has a Station.
- ☐ Every operation can point to a separate Repair Station.
- ☐ Station types match the way work will be performed.

Create Shifts

Define when production runs and make Shifts available for Work Order assignment.

<https://www.fynd.com/cookbooks/forge/configure-factory/shifts>

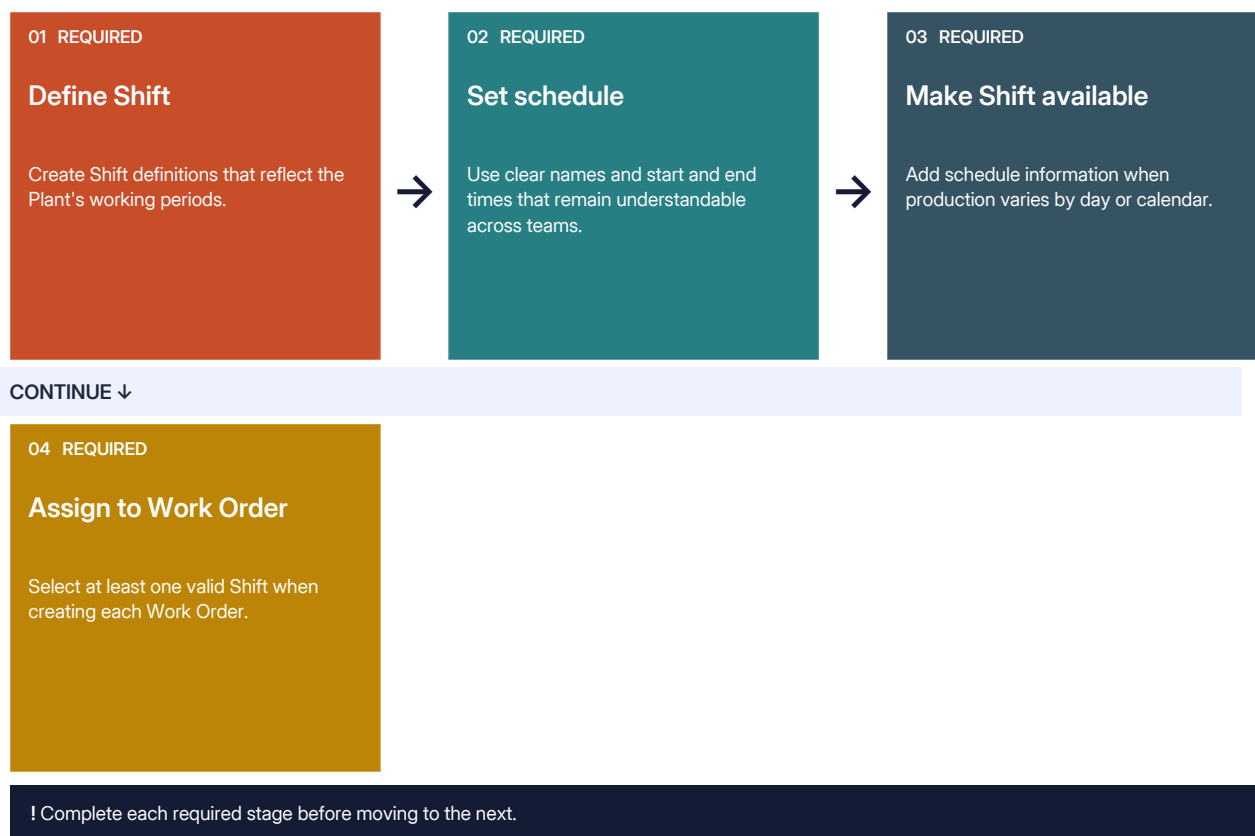
Requirement
Required

Before you begin

- The Forge Factory Model

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Work Order cannot be created or started without at least one Shift assignment.

What to do

- 1 Create Shift definitions that reflect the Plant's working periods.
- 2 Use clear names and start and end times that remain understandable across teams.
- 3 Add schedule information when production varies by day or calendar.
- 4 Select at least one valid Shift when creating each Work Order.

Rules to remember

- A Work Order cannot be created or started without at least one Shift assignment.
- Assigned Shifts must belong to the same Plant.
- Shift information can be used in sequencing and identifier templates.

Ready when

- [] All active production periods are represented.
- [] Shift times do not overlap unintentionally.
- [] At least one Shift is ready for the first Work Order.

Factory Readiness Checklist

Confirm the physical factory model is complete before creating production master data.

<https://www.fynd.com/cookbooks/forge/configure-factory/factory-readiness>

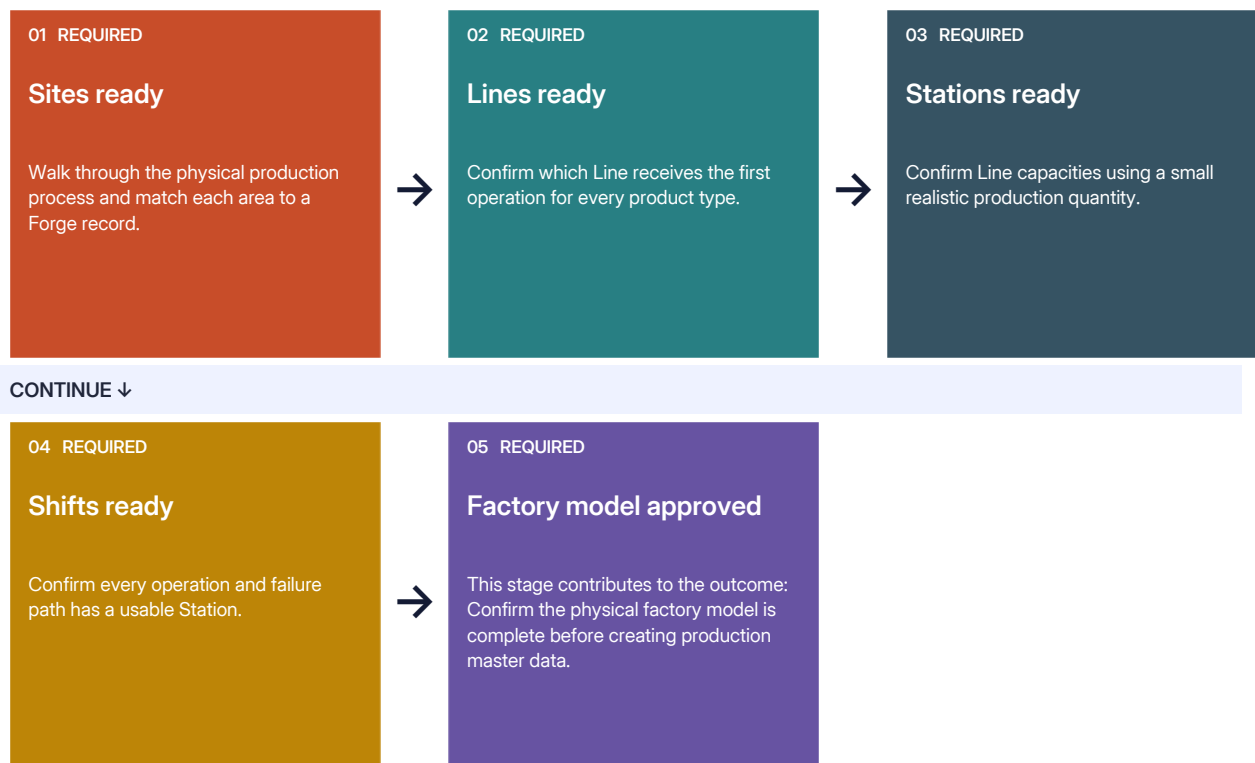
Requirement	Required
-------------	----------

Before you begin

- Create Sites
- Create Production Lines
- Create Stations and Repair Stations
- Create Shifts

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



! Complete each required stage before moving to the next.

Why this connection matters: Do not compensate for missing Lines by creating extra Stations.

What to do

- 1 Walk through the physical production process and match each area to a Forge record.
- 2 Confirm which Line receives the first operation for every product type.

- 3 Confirm Line capacities using a small realistic production quantity.
- 4 Confirm every operation and failure path has a usable Station.

Rules to remember

- Do not compensate for missing Lines by creating extra Stations.
- Do not mark every Line as a Starting Line unless units can genuinely begin there.
- Repair Stations must be planned before Routings are built.

Ready when

- ☐ Required Sites exist.
- ☐ Lines are active, correctly typed, and have explicit capacity.
- ☐ Initial Lines are marked as Starting Lines.
- ☐ Operation and Repair Stations exist at valid Sites.
- ☐ At least one valid Shift exists.

GUIDED STAGE

Define Products and Materials

Organise products, create components and variants, build BOMs, and choose the correct material tracking approach.

<https://www.fynd.com/cookbooks/forge/define-products>

Follow this stage in order

- 1 [Projects and Product Families](#) - Use the recommended hierarchy to keep products, reporting, and future growth organised.
- 2 [Create Components](#) - Create the material and part records that will be used in Product BOMs and shop-floor validation.
- 3 [Create Products and Variants](#) - Create finished-product definitions and decide when Variants are useful.
- 4 [Build the Bill of Materials](#) - Define exactly which Components and quantities are required to build the Product.
- 5 [Activate a BOM Version](#) - Move the reviewed BOM from draft to the active production version required by Routing.
- 6 [Serial, Lot, and Mixed Material Tracking](#) - Choose how components are traced while keeping finished production aligned with Forge's unit-serial model.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Projects and Product Families

Use the recommended hierarchy to keep products, reporting, and future growth organised.

<https://www.fynd.com/cookbooks/forge/define-products/projects-and-product-families>

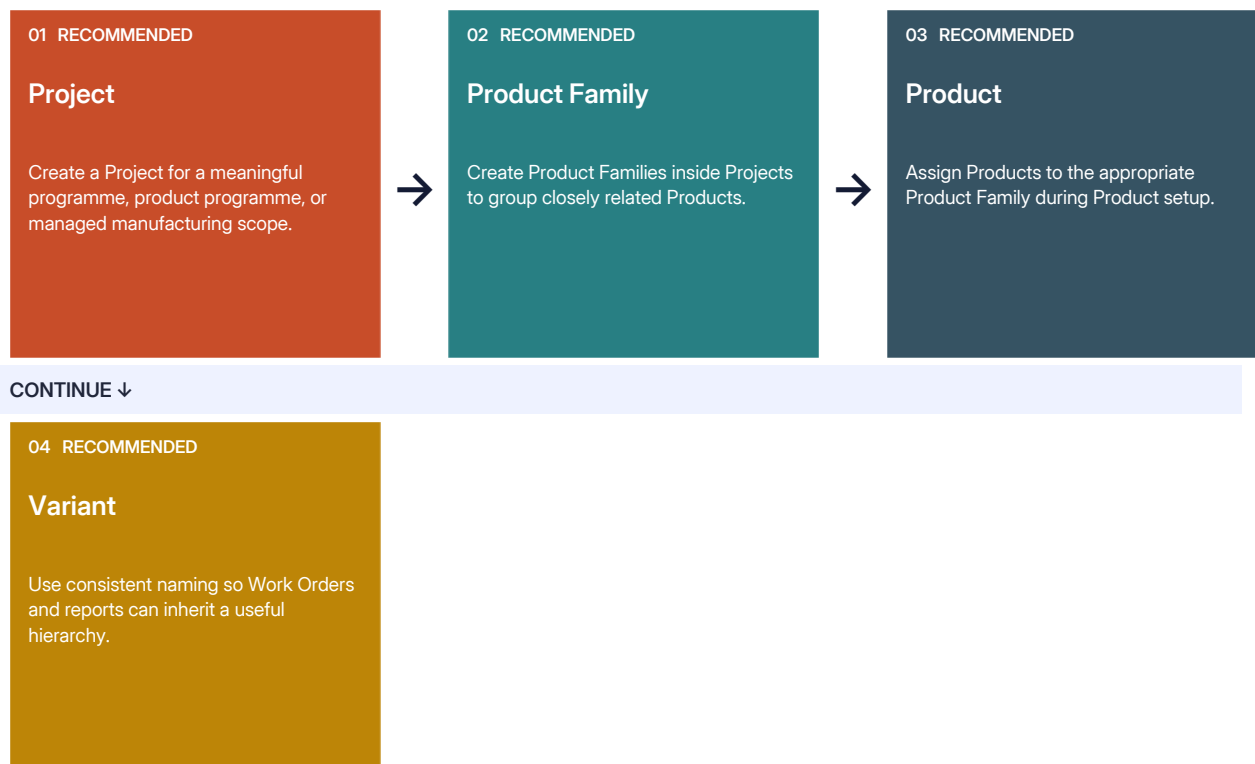
Requirement	Recommended
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Before you begin

- Factory Readiness Checklist

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



! Complete each required stage before moving to the next.

Why this connection matters: A Product Family requires a Project.

What to do

- 1 Create a Project for a meaningful programme, product programme, or managed manufacturing scope.
- 2 Create Product Families inside Projects to group closely related Products.
- 3 Assign Products to the appropriate Product Family during Product setup.
- 4 Use consistent naming so Work Orders and reports can inherit a useful hierarchy.

Rules to remember

- A Product Family requires a Project.
- Forge can create a Product without a Product Family, but the hierarchy improves organisation and reporting.
- Use Variants for versions of the same Product, not for unrelated Products.

Ready when

- ☐ Projects have a clear purpose.
- ☐ Product Families group genuinely related Products.
- ☐ The hierarchy will remain understandable as more Products are added.

Create Components

Create the material and part records that will be used in Product BOMs and shop-floor validation.

<https://www.fynd.com/cookbooks/forge/define-products/components>

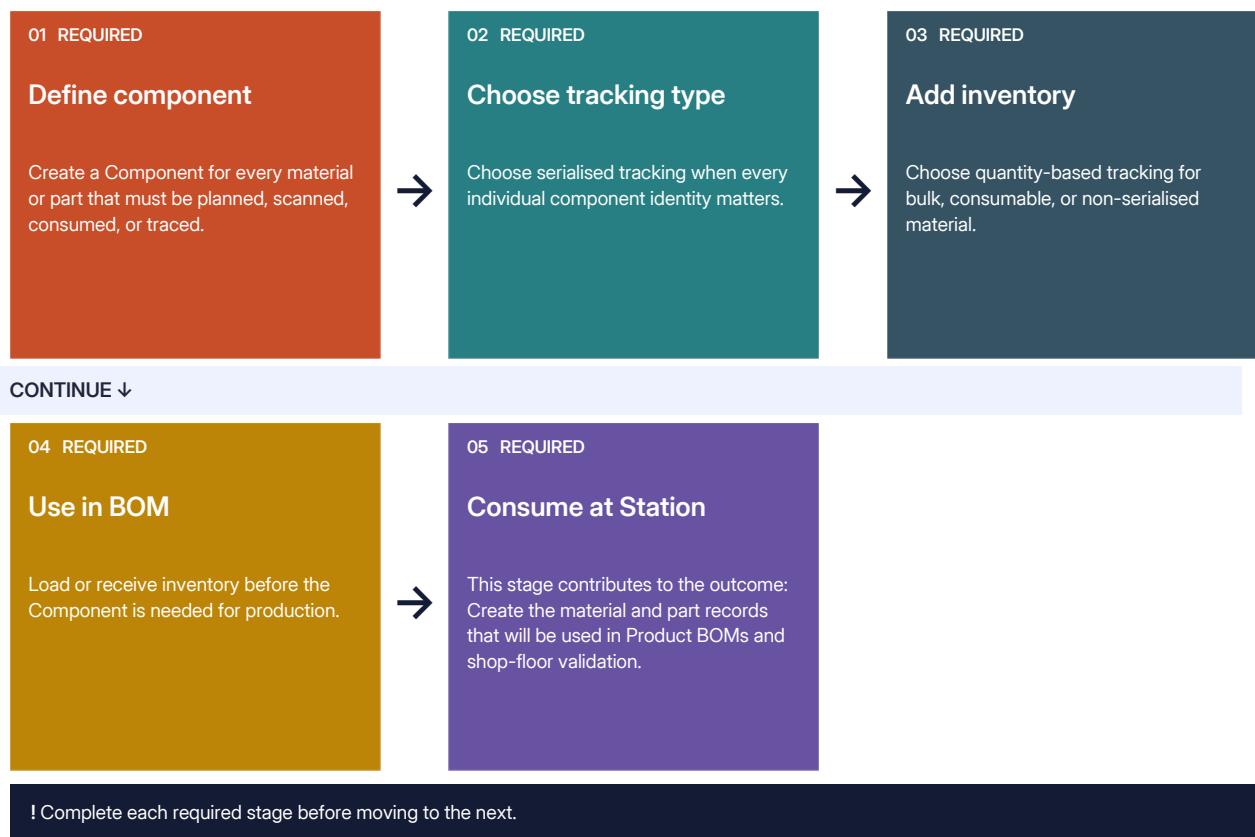
Requirement	Required
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Before you begin

- Projects and Product Families

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: BOMs reference Component records, so Components must exist first.

What to do

- 1 Create a Component for every material or part that must be planned, scanned, consumed, or traced.
- 2 Choose serialised tracking when every individual component identity matters.
- 3 Choose quantity-based tracking for bulk, consumable, or non-serialised material.
- 4 Load or receive inventory before the Component is needed for production.

Rules to remember

- BOMs reference Component records, so Components must exist first.
- Serialised Components require a valid individual identifier during consumption.
- Quantity-based Components are validated against the required BOM quantity.

Ready when

- ☐ Every BOM material has a Component record.
- ☐ Tracking type matches the physical identification method.
- ☐ Units of measure and quantities are consistent.

Create Products and Variants

Create finished-product definitions and decide when Variants are useful.

<https://www.fynd.com/cookbooks/forge/define-products/products-and-variants>

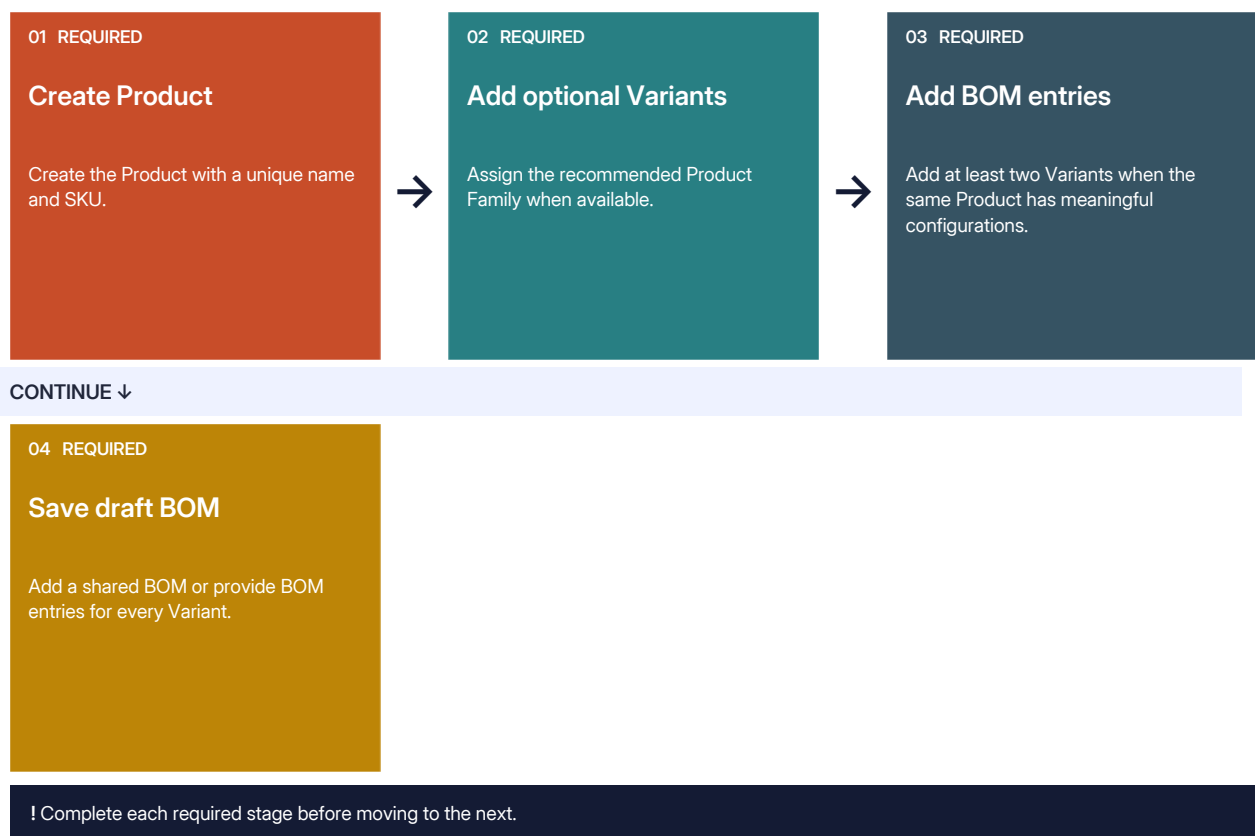
Requirement	Required
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Before you begin

- Create Components

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Product without Variants requires a shared BOM.

What to do

- 1 Create the Product with a unique name and SKU.
- 2 Assign the recommended Product Family when available.
- 3 Add at least two Variants when the same Product has meaningful configurations.
- 4 Add a shared BOM or provide BOM entries for every Variant.
- 5 Configure a serial number template when a specific finished-unit format is needed.

Rules to remember

- A Product without Variants requires a shared BOM.
- A Product with Variants needs a shared BOM or a BOM for every Variant.
- If Variants are used, a Work Order must select the required Variant.
- The serial number template is optional, but unit serial generation before release is not.

Ready when

- ☐ Product name and SKU are unique.
- ☐ Variant use is deliberate and consistent.
- ☐ Every Product configuration has BOM coverage.

Build the Bill of Materials

Define exactly which Components and quantities are required to build the Product.

<https://www.fynd.com/cookbooks/forge/define-products/bill-of-materials>

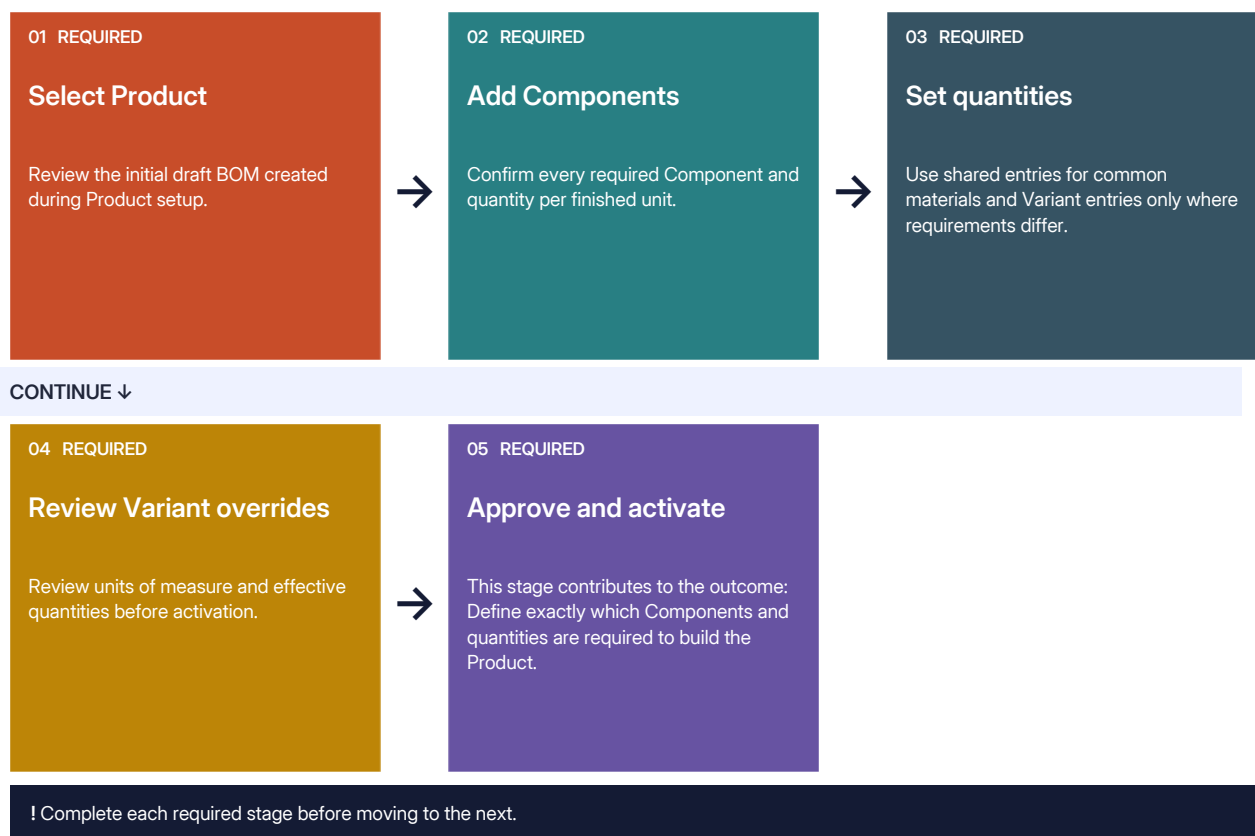
Requirement	Required
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Before you begin

- Create Products and Variants

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Routing creation requires an active BOM belonging to the selected Product.

What to do

- 1 Review the initial draft BOM created during Product setup.
- 2 Confirm every required Component and quantity per finished unit.
- 3 Use shared entries for common materials and Variant entries only where requirements differ.
- 4 Review units of measure and effective quantities before activation.

Rules to remember

- Routing creation requires an active BOM belonging to the selected Product.
- The pre-start readiness check uses the active BOM to calculate material requirements.
- Changing an active BOM can change future production readiness and should be controlled.

Ready when

- ☐ The BOM contains every required production Component.
- ☐ Quantities match one finished unit.
- ☐ Variant overrides do not duplicate shared requirements incorrectly.

Activate a BOM Version

Move the reviewed BOM from draft to the active production version required by Routing.

<https://www.fynd.com/cookbooks/forge/define-products/activate-bom>

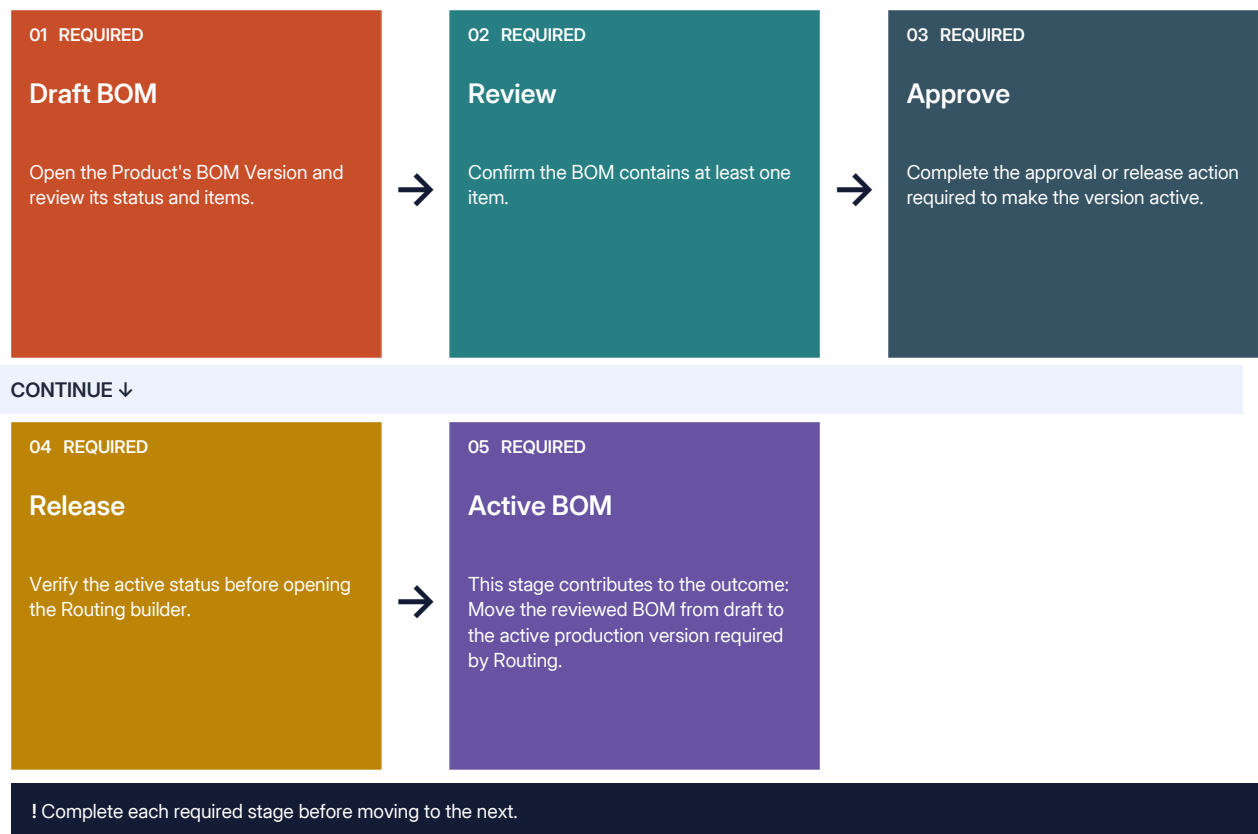
Requirement	Required
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Before you begin

- Build the Bill of Materials

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Only reviewed draft or approved versions should be released.

What to do

- 1 Open the Product's BOM Version and review its status and items.
- 2 Confirm the BOM contains at least one item.
- 3 Complete the approval or release action required to make the version active.
- 4 Verify the active status before opening the Routing builder.

Rules to remember

- Only reviewed draft or approved versions should be released.
- An empty BOM cannot be released as a meaningful production definition.
- Routing must use an active BOM from the same Product.

Ready when

- ☐ The correct BOM Version is active.
- ☐ The active BOM belongs to the intended Product.
- ☐ Material inventory exists or has a replenishment plan.

Serial, Lot, and Mixed Material Tracking

Choose how components are traced while keeping finished production aligned with Forge's unit-serial model.

<https://www.fynd.com/cookbooks/forge/define-products/material-tracking>

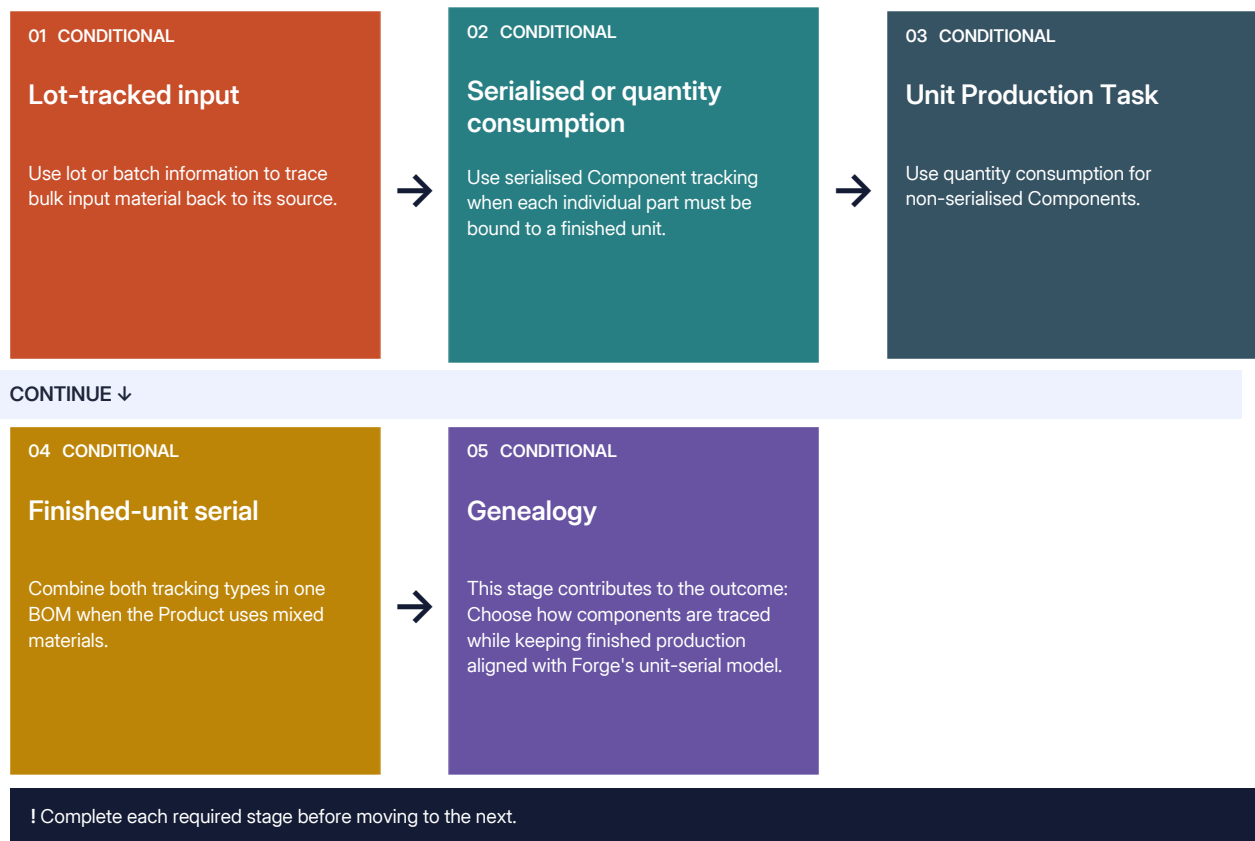
Requirement	Conditional
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Before you begin

- Create Components
- Build the Bill of Materials

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Normal Forge Work Orders still create one finished-unit serial and Production Task per output unit.

What to do

- 1 Use lot or batch information to trace bulk input material back to its source.
- 2 Use serialised Component tracking when each individual part must be bound to a finished unit.

- 3 Use quantity consumption for non-serialised Components.
- 4 Combine both tracking types in one BOM when the Product uses mixed materials.

Rules to remember

- Normal Forge Work Orders still create one finished-unit serial and Production Task per output unit.
- A serial number batch groups finished-unit serials; it is not a quantity-only production mode.
- Material genealogy is strongest when lots, serials, and consumption events are captured at the correct Station.

Ready when

- ☐ Every Component has the correct tracking type.
- ☐ Lot identifiers are available for batch-tracked inputs.
- ☐ The team understands that finished output remains unit-serialised.

GUIDED STAGE

Design the Production Process

Turn the product definition into an executable route, connect Stations, assign Lines, and prepare identifiers.

<https://www.fynd.com/cookbooks/forge/design-process>

Follow this stage in order

- 1 [Routing in Plain Language](#) - Understand how a Routing tells Forge where a unit goes and what happens at each operation.
- 2 [Create and Validate a Routing](#) - Build a connected production workflow that passes Forge's Routing validation.
- 3 [Assign Products to Lines](#) - Tell Forge which active Lines are allowed to receive tasks for a Product.
- 4 [Starting Lines and Capacity](#) - Configure where newly released units enter production and how Forge prevents Line overload.
- 5 [Configure Product Identifiers](#) - Prepare serial, range, template, or bundle identifiers that must be assigned when Production Tasks are created.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Routing in Plain Language

Understand how a Routing tells Forge where a unit goes and what happens at each operation.

<https://www.fynd.com/cookbooks/forge/design-process/routing-basics>

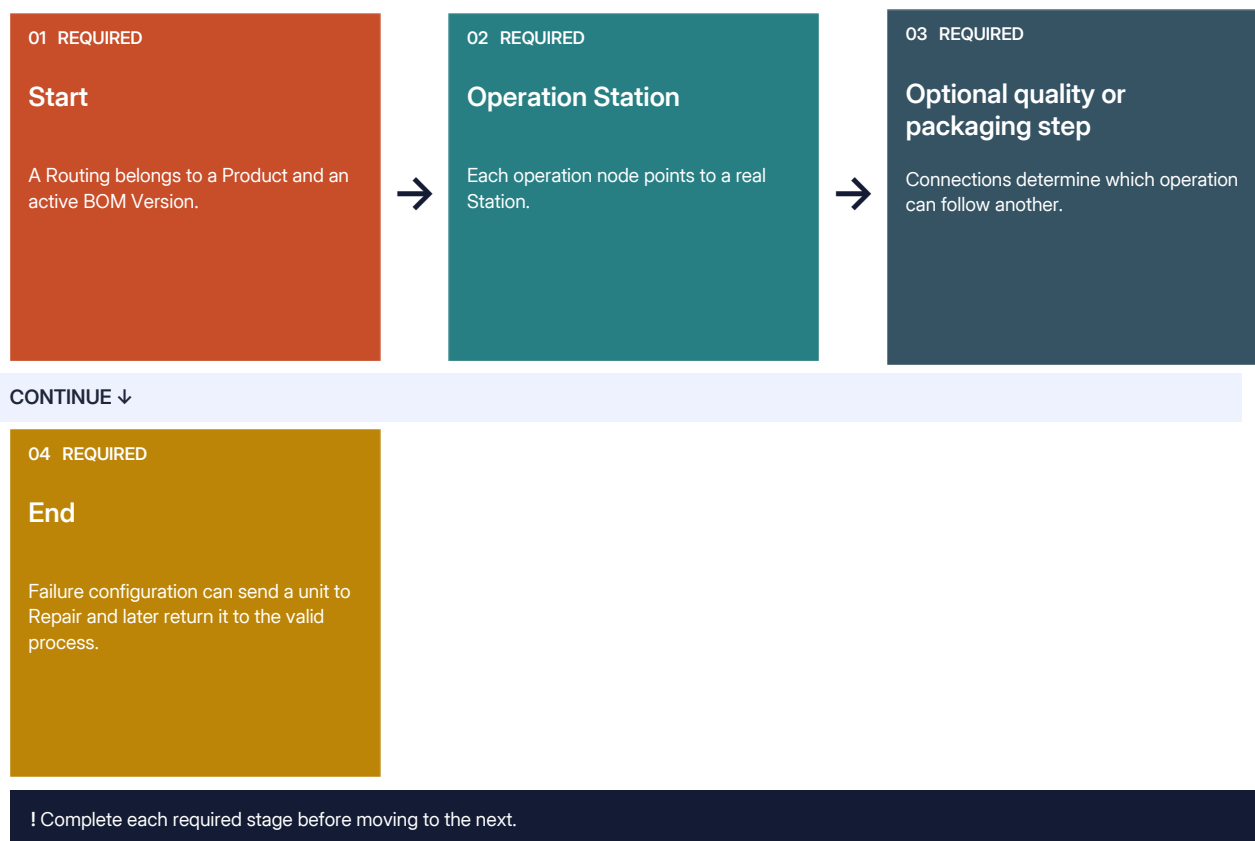
Requirement	Required
-------------	----------

Before you begin

- Activate a BOM Version
- Create Stations and Repair Stations

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Work Order cannot be created for a Product with no Routing.

What to do

- 1 A Routing belongs to a Product and an active BOM Version.
- 2 Each operation node points to a real Station.
- 3 Connections determine which operation can follow another.

- 4 Failure configuration can send a unit to Repair and later return it to the valid process.

Rules to remember

- A Work Order cannot be created for a Product with no Routing.
- Only active Routings are offered in the normal Work Order start journey.
- Components and tools can be attached to the operation where they are needed.

Ready when

- ☐ I can explain the route from Start to End.
- ☐ Every operation maps to a real Station.
- ☐ Failure paths have an intentional destination.

Create and Validate a Routing

Build a connected production workflow that passes Forge's Routing validation.

<https://www.fynd.com/cookbooks/forge/design-process/create-routing>

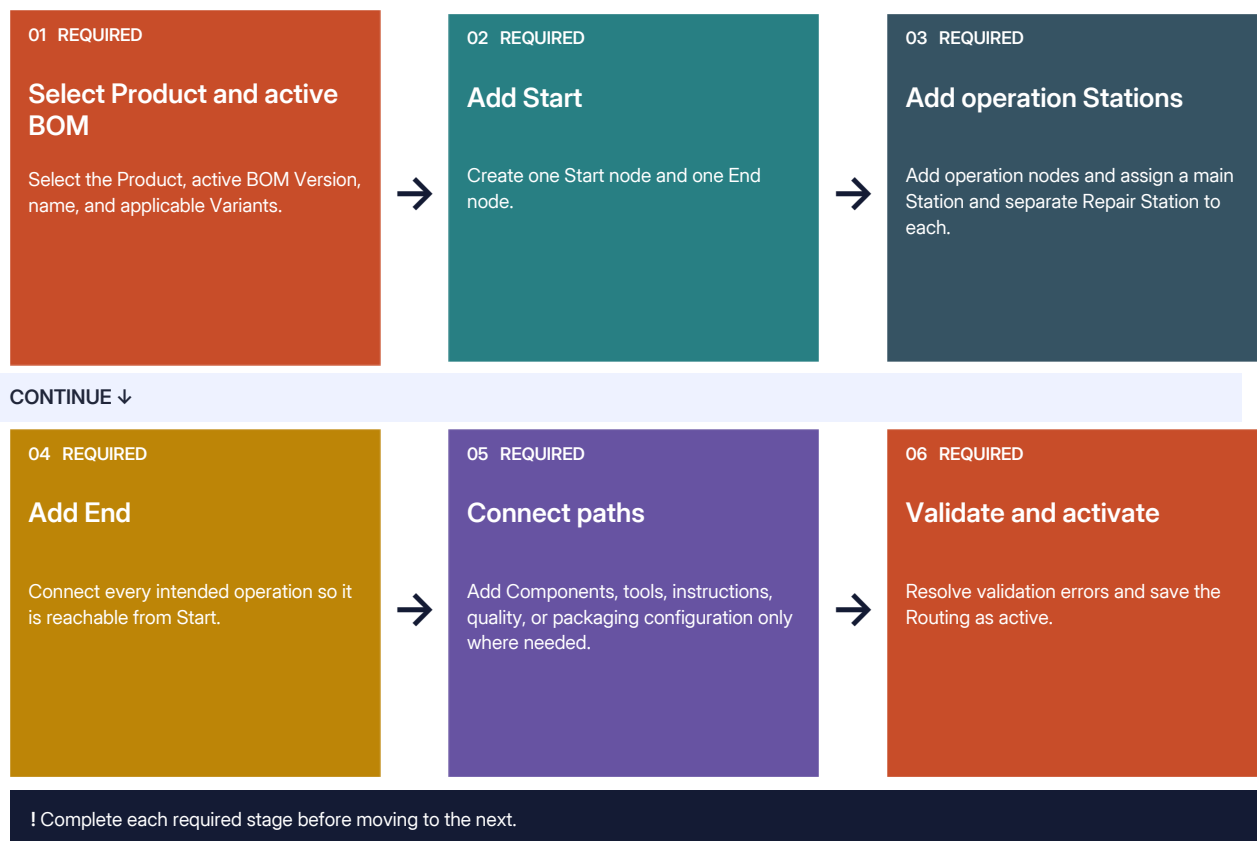
Requirement	Required
-------------	----------

Before you begin

- Routing in Plain Language

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Routing needs at least one operation Station.

What to do

- 1 Select the Product, active BOM Version, name, and applicable Variants.
- 2 Create one Start node and one End node.
- 3 Add operation nodes and assign a main Station and separate Repair Station to each.
- 4 Connect every intended operation so it is reachable from Start.

- 5 Add Components, tools, instructions, quality, or packaging configuration only where needed.
- 6 Resolve validation errors and save the Routing as active.

Rules to remember

- A Routing needs at least one operation Station.
- Every operation requires a Repair Station different from its main Station.
- Disconnected operations are not a valid executable process.
- Loops are allowed only when they have a clear exit condition.

Ready when

- [] Start and End are present.
- [] Every operation is reachable from Start.
- [] Every operation has valid main and Repair Stations.
- [] The Routing is active and uses the correct active BOM.

Assign Products to Lines

Tell Forge which active Lines are allowed to receive tasks for a Product.

<https://www.fynd.com/cookbooks/forge/design-process/product-line-assignment>

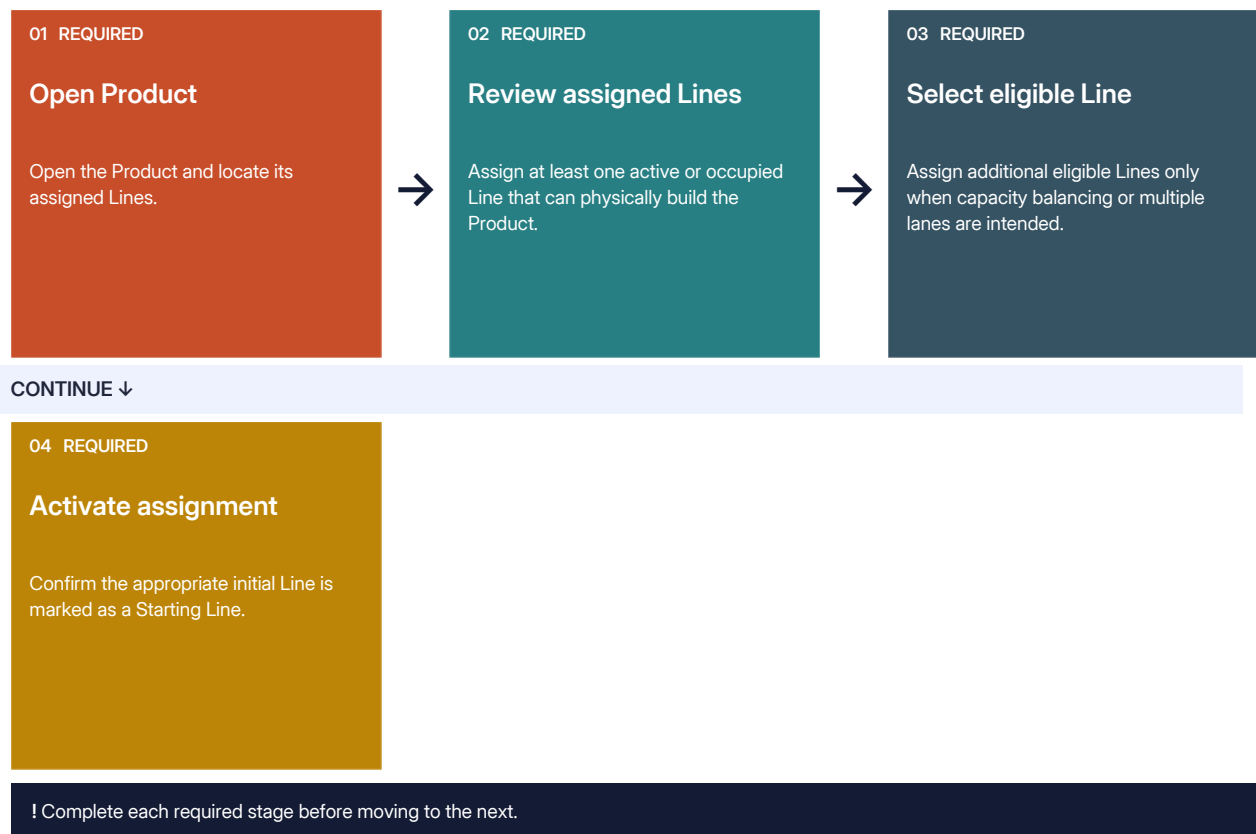
Requirement	Required
-------------	----------

Before you begin

- Create Production Lines
- Create Products and Variants

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Work Order creation is blocked when the Product has no active Line assignment.

What to do

- 1 Open the Product and locate its assigned Lines.
- 2 Assign at least one active or occupied Line that can physically build the Product.
- 3 Assign additional eligible Lines only when capacity balancing or multiple lanes are intended.
- 4 Confirm the appropriate initial Line is marked as a Starting Line.

Rules to remember

- Work Order creation is blocked when the Product has no active Line assignment.
- The current assignment flow allows one active Product assignment per Line.
- Removing an assignment can be blocked while active Production Tasks use that Line.

Ready when

- ☐ The Product has at least one active Line.
- ☐ Assigned Lines match the physical process.
- ☐ At least one assigned initial Line is a Starting Line.

Starting Lines and Capacity

Configure where newly released units enter production and how Forge prevents Line overload.

<https://www.fynd.com/cookbooks/forge/design-process/starting-lines-and-capacity>

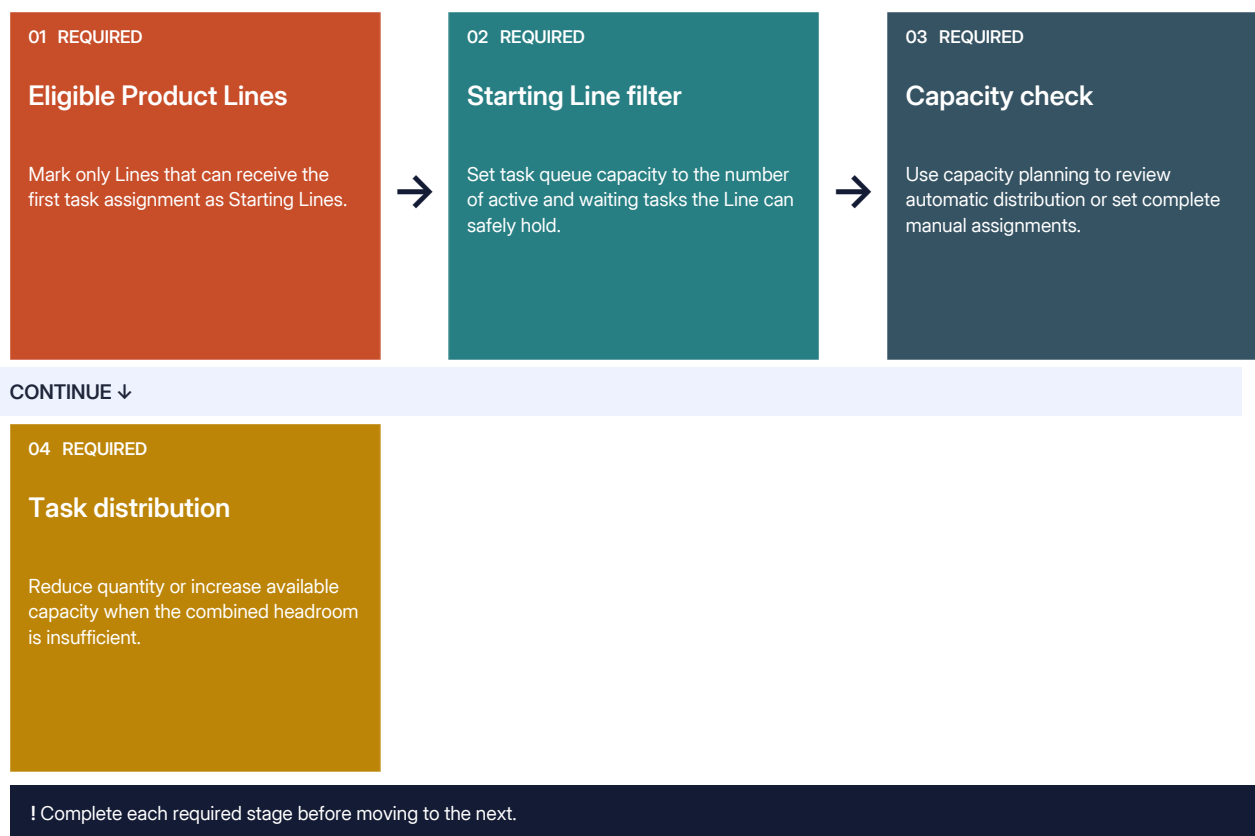
Requirement	Required
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Before you begin

- Assign Products to Lines

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Initial Work Orders normally require an assigned active Starting Line.

What to do

- 1 Mark only Lines that can receive the first task assignment as Starting Lines.
- 2 Set task queue capacity to the number of active and waiting tasks the Line can safely hold.
- 3 Use capacity planning to review automatic distribution or set complete manual assignments.
- 4 Reduce quantity or increase available capacity when the combined headroom is insufficient.

Rules to remember

- Initial Work Orders normally require an assigned active Starting Line.
- Downstream Lines receive tasks as units progress; they should not all be Starting Lines.
- Manual capacity planning must cover every task sequence without gaps.

Ready when

- [] Starting Lines reflect the real first manufacturing phase.
- [] Queue capacity is explicit and realistic.
- [] Combined headroom can accept the trial Work Order quantity.

Configure Product Identifiers

Prepare serial, range, template, or bundle identifiers that must be assigned when Production Tasks are created.

<https://www.fynd.com/cookbooks/forge/design-process/product-identifiers>

Requirement	Conditional
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Before you begin

- Create Products and Variants

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Required task-creation mappings block release when availability is insufficient.

What to do

- 1 Create an identifier mapping for each identity that must be attached to a finished unit.
- 2 Choose a template when identifiers can be generated dynamically.
- 3 Choose ranges or bundles when identifiers come from controlled preloaded values.
- 4 Ensure required mappings have at least one available value for every Work Order unit.

Rules to remember

- Required task-creation mappings block release when availability is insufficient.
- Optional identifier shortages appear as warnings.
- The finished-unit serial batch is still required even when additional identifiers are configured.

Ready when

- ☐ Every required identifier mapping has a valid source.
- ☐ Available quantity covers the Work Order.
- ☐ Templates and ranges generate the intended format.

GUIDED STAGE

Plan and Launch Production

Create production demand, generate unit serials, plan Line capacity, validate readiness, and release work safely.

<https://www.fynd.com/cookbooks/forge/plan-production>

Follow this stage in order

- 1 [Production Order vs Work Order](#) - Choose the right planning record and understand which one creates executable production.
- 2 [Create a Work Order](#) - Create an executable production instruction after all Product and factory dependencies are ready.
- 3 [Generate and Confirm Unit Serials](#) - Create exactly one finished-unit serial for every unit before releasing a normal Work Order.
- 4 [Plan Capacity and Select Routing](#) - Confirm eligible Starting Lines have enough headroom and choose the active Product Routing.
- 5 [Pre-Start Validation and Release](#) - Use Forge's readiness checks to fix blockers before Production Tasks are created.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Production Order vs Work Order

Choose the right planning record and understand which one creates executable production.

<https://www.fynd.com/cookbooks/forge/plan-production/production-order-vs-work-order>

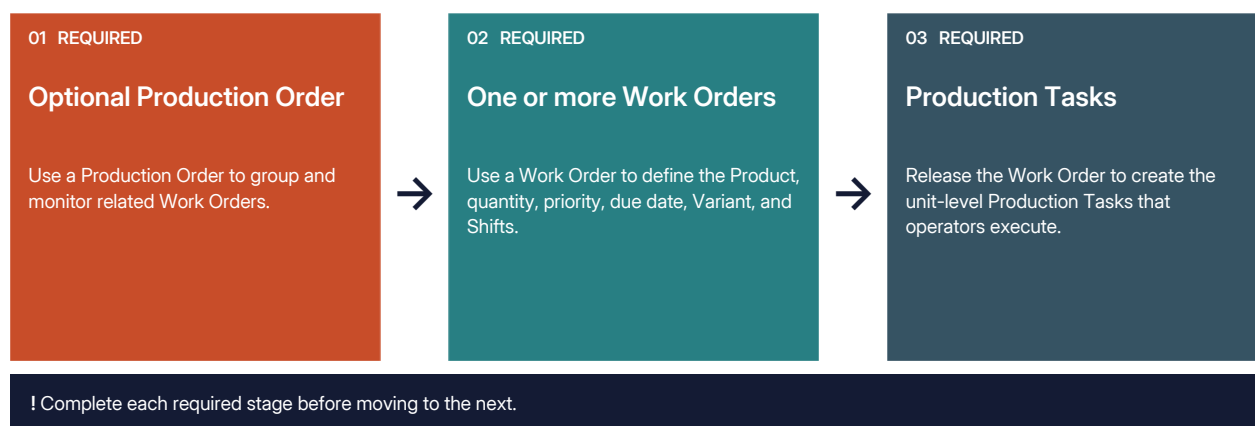
Requirement	Required
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Before you begin

- Create and Validate a Routing
- Assign Products to Lines
- Create Shifts

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Production Order is optional and does not start production by itself.

What to do

- 1 Use a Production Order to group and monitor related Work Orders.
- 2 Use a Work Order to define the Product, quantity, priority, due date, Variant, and Shifts.
- 3 Release the Work Order to create the unit-level Production Tasks that operators execute.

Rules to remember

- A Production Order is optional and does not start production by itself.
- A Work Order can exist without being grouped under a Production Order.
- Production Order completion follows the completion of its Work Orders.

Ready when

- [] The planning record matches the business need.

- [] Every executable quantity is represented by a Work Order.
- [] Grouping is used only where it improves control.

Create a Work Order

Create an executable production instruction after all Product and factory dependencies are ready.

<https://www.fynd.com/cookbooks/forge/plan-production/create-work-order>

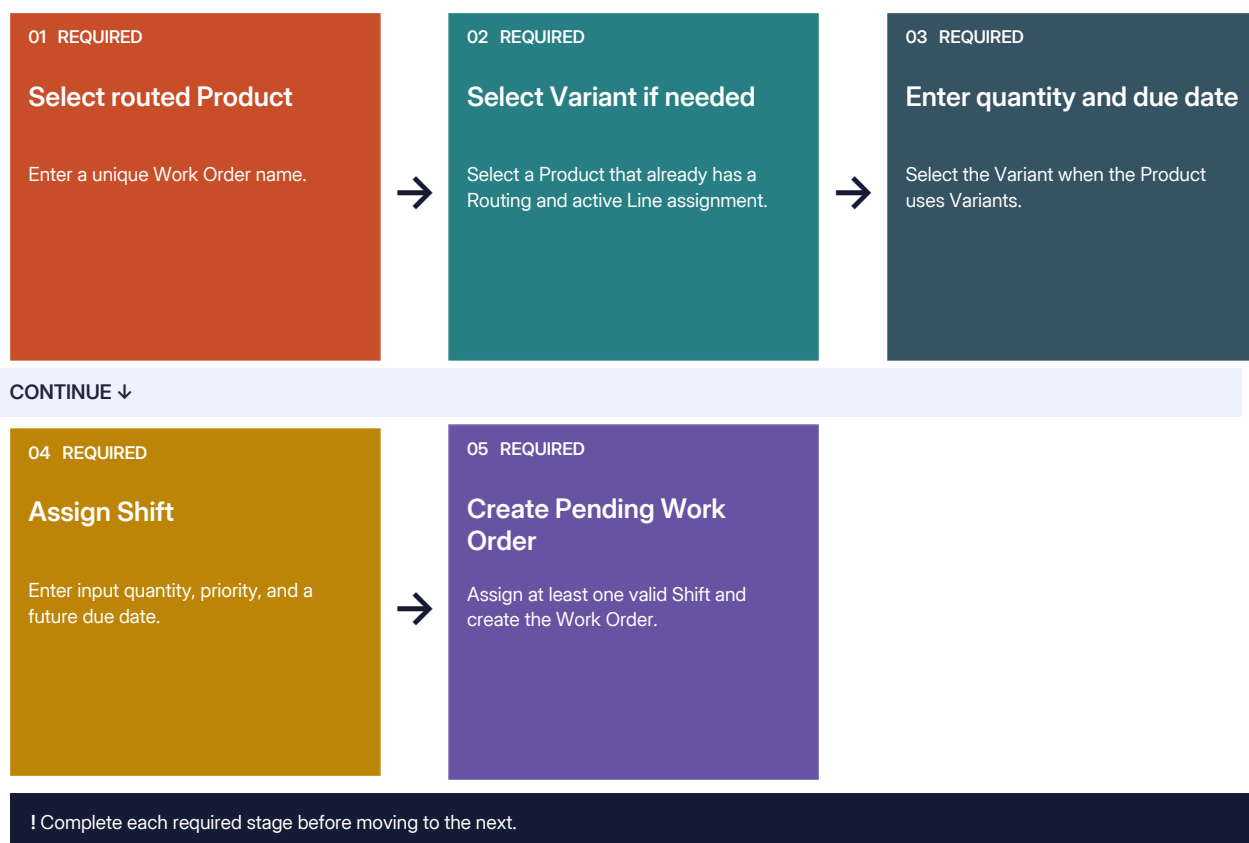
Requirement	Required
-------------	----------

Before you begin

- Production Order vs Work Order
- Create and Validate a Routing
- Assign Products to Lines
- Create Shifts

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Quantity must be at least one.

What to do

- 1 Enter a unique Work Order name.

- 2 Select a Product that already has a Routing and active Line assignment.
- 3 Select the Variant when the Product uses Variants.
- 4 Enter input quantity, priority, and a future due date.
- 5 Assign at least one valid Shift and create the Work Order.

Rules to remember

- Quantity must be at least one.
- The Product multiplier can increase the final unit quantity.
- A Product with no Routing or active Line assignment is rejected.
- At least one Shift is mandatory.

Ready when

- ☐ Product, Variant, quantity, priority, and due date are correct.
- ☐ At least one Shift is selected.
- ☐ The resulting final quantity is understood.

Generate and Confirm Unit Serials

Create exactly one finished-unit serial for every unit before releasing a normal Work Order.

<https://www.fynd.com/cookbooks/forge/plan-production/generate-unit-serials>

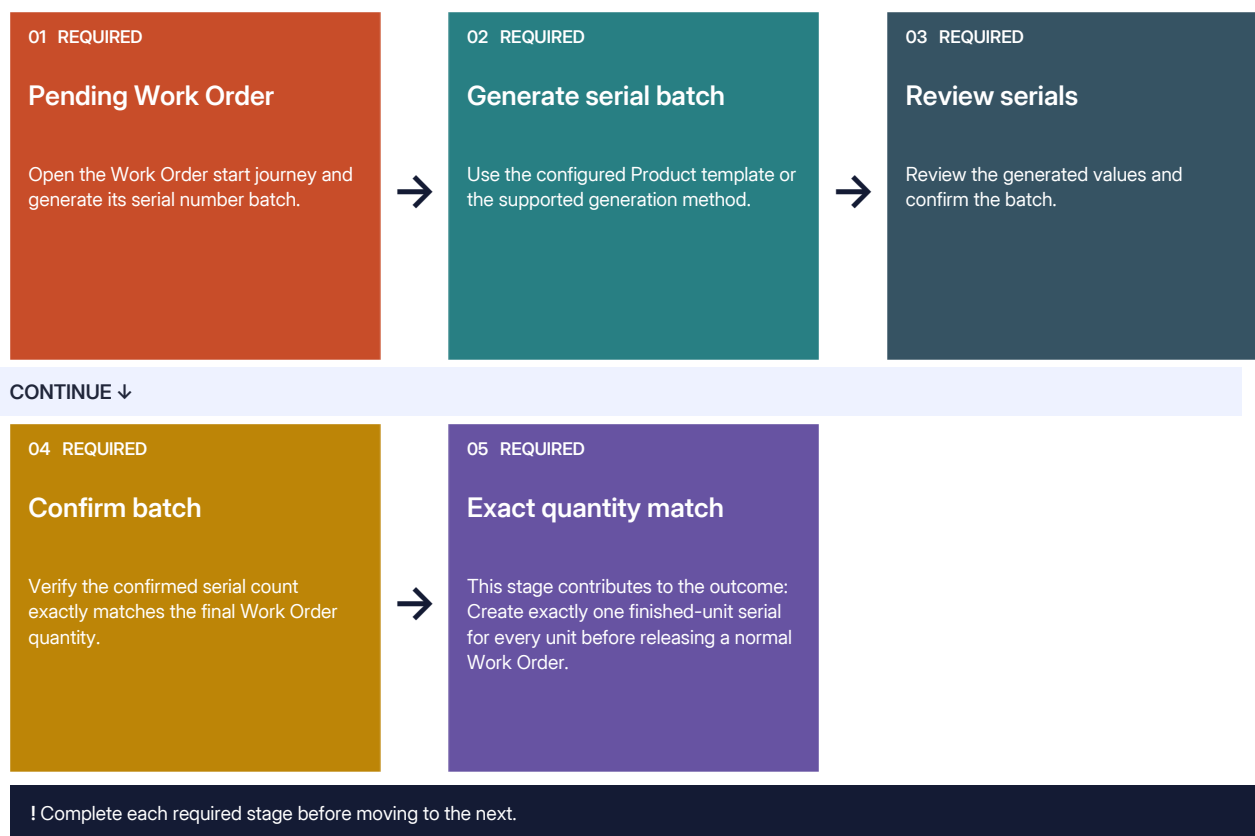
Requirement	Required
-------------	----------

Before you begin

- Create a Work Order

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: No normal Work Order can start without a serial number batch.

What to do

- 1 Open the Work Order start journey and generate its serial number batch.
- 2 Use the configured Product template or the supported generation method.
- 3 Review the generated values and confirm the batch.
- 4 Verify the confirmed serial count exactly matches the final Work Order quantity.

Rules to remember

- No normal Work Order can start without a serial number batch.
- The batch must be confirmed.
- Serial count must exactly equal Work Order quantity.
- A serial number batch groups unit serials; it does not replace unit-level identities.

Ready when

- ☐ The serial batch is confirmed.
- ☐ Serial count equals final quantity.
- ☐ Serial format is suitable for scanning and traceability.

Plan Capacity and Select Routing

Confirm eligible Starting Lines have enough headroom and choose the active Product Routing.

<https://www.fynd.com/cookbooks/forge/plan-production/capacity-and-routing>

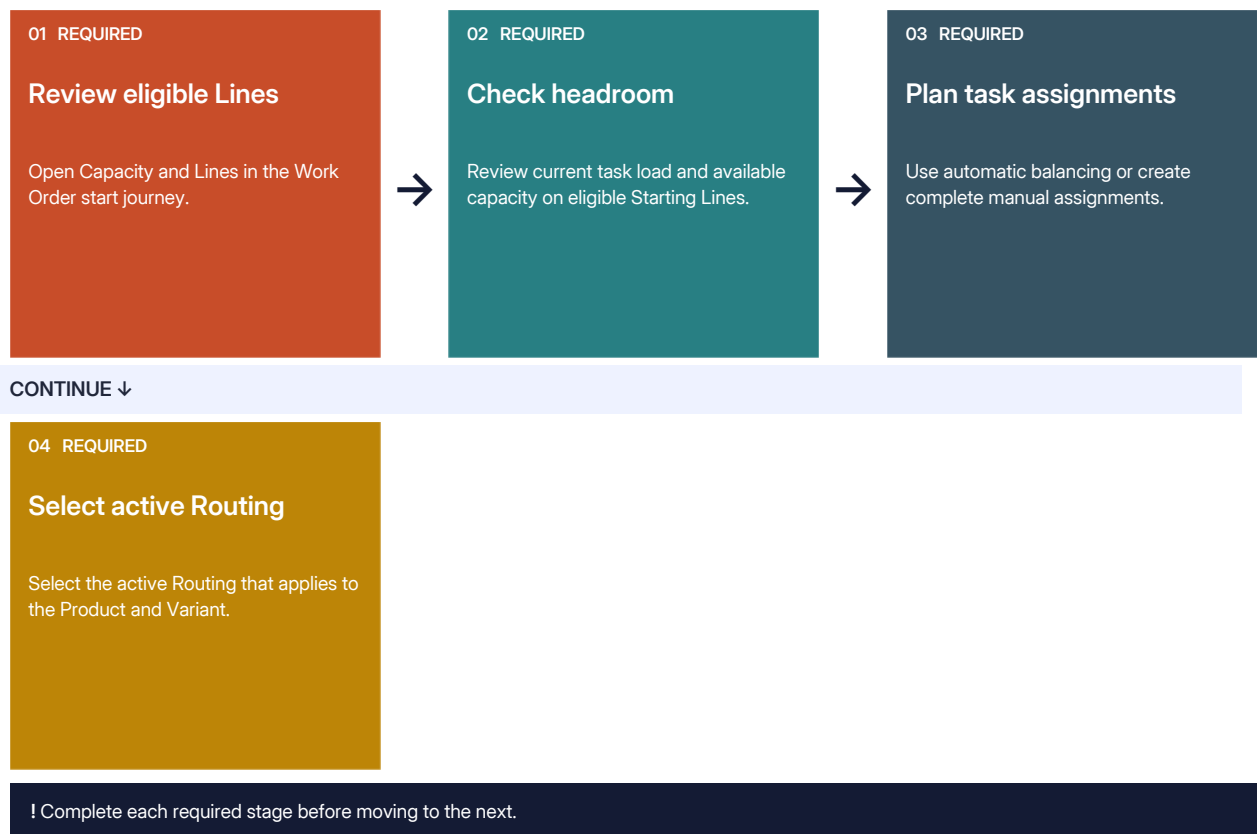
Requirement	Required
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Before you begin

- Generate and Confirm Unit Serials
- Starting Lines and Capacity

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Total available queue capacity must cover the Work Order quantity.

What to do

- 1 Open Capacity and Lines in the Work Order start journey.
- 2 Review current task load and available capacity on eligible Starting Lines.
- 3 Use automatic balancing or create complete manual assignments.
- 4 Select the active Routing that applies to the Product and Variant.

Rules to remember

- Total available queue capacity must cover the Work Order quantity.
- Every manually assigned task must reference an eligible Line.
- The selected Routing must be active and applicable to the Product configuration.

Ready when

- ☐ Capacity planning is complete.
- ☐ No Line exceeds its queue capacity.
- ☐ The correct active Routing is selected.

Pre-Start Validation and Release

Use Forge's readiness checks to fix blockers before Production Tasks are created.

<https://www.fynd.com/cookbooks/forge/plan-production/pre-start-validation>

Requirement	Required
-------------	----------

Before you begin

- Plan Capacity and Select Routing

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A missing active BOM is a blocker in the supported release journey.

What to do

- Run pre-start validation after serials, capacity, and Routing are ready.
- Resolve every blocker, beginning with Product, active BOM, serials, and global material shortages.
- Transfer material to the first Station when first-stage availability is insufficient.
- Review later-Station shortages and warnings before deciding whether to proceed.

- 5 Release the Work Order only when the result allows production to start.

Rules to remember

- A missing active BOM is a blocker in the supported release journey.
- Insufficient first-stage material blocks release.
- Later-stage shortages can be warnings because material may arrive before the unit reaches that Station.
- Required identifier shortages block release.

Ready when

- ☐ All blockers are resolved.
- ☐ Warnings have an understood mitigation.
- ☐ The Work Order changes to In Progress and creates the expected number of Production Tasks.

GUIDED STAGE

Run Production

Understand Production Tasks, operator execution, material consumption, and the status trail created during work.

<https://www.fynd.com/cookbooks/forge/run-production>

Follow this stage in order

- 1 [Production Tasks](#) - Understand the unit-level record operators execute after a Work Order is released.
- 2 [Operator Execution Flow](#) - Follow the normal sequence an operator uses to process a unit at a Station.
- 3 [Material Consumption During Production](#) - Capture the material used by each unit so inventory and genealogy remain accurate.
- 4 [Statuses and Route Logs](#) - Read the history Forge creates as Work Orders and units move through production.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Production Tasks

Understand the unit-level record operators execute after a Work Order is released.

<https://www.fynd.com/cookbooks/forge/run-production/production-tasks>

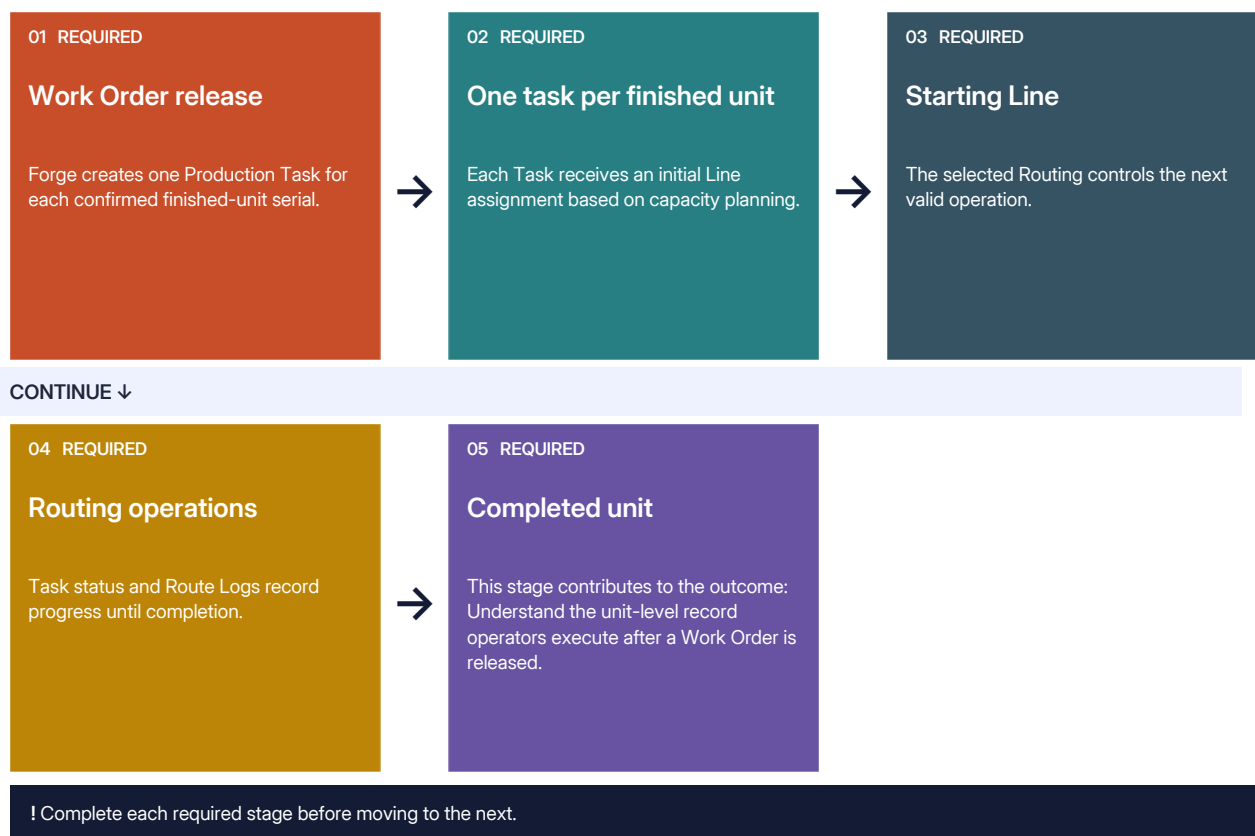
Requirement	Required
-------------	----------

Before you begin

- Pre-Start Validation and Release

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Production Tasks require a Line assignment.

What to do

- 1 Forge creates one Production Task for each confirmed finished-unit serial.
- 2 Each Task receives an initial Line assignment based on capacity planning.
- 3 The selected Routing controls the next valid operation.
- 4 Task status and Route Logs record progress until completion.

Rules to remember

- Production Tasks require a Line assignment.
- A Task cannot skip an operation unless Routing or configured auto-pass behaviour allows it.
- The live Task Line can change as the unit moves through production.

Ready when

- [] Task count matches Work Order quantity.
- [] Every Task has a serial and eligible Line.
- [] Operators can find Tasks at the correct Station.

Operator Execution Flow

Follow the normal sequence an operator uses to process a unit at a Station.

<https://www.fynd.com/cookbooks/forg/run-production/operator-flow>

Requirement	Required
-------------	----------

Before you begin

- Production Tasks

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



! Complete each required stage before moving to the next.

Why this connection matters: Operators should work from the Station that matches the current Routing step.

What to do

- 1 Open the Station work area and identify or scan the unit.
- 2 Review the current operation, instructions, required inputs, and quality gates.
- 3 Scan serialised Components or enter the exact quantity for quantity-based Components.
- 4 Complete required checks and record the operation result.
- 5 Finish the step so Forge can move the unit to the next valid operation.

Rules to remember

- Operators should work from the Station that matches the current Routing step.
- Required Component and verification inputs block completion when missing.
- Failed results must follow the configured exception path.

Ready when

- [] The correct unit and operation are visible.
- [] Required inputs validate successfully.
- [] The completed step appears in the unit history.

Material Consumption During Production

Capture the material used by each unit so inventory and genealogy remain accurate.

<https://www.fynd.com/cookbooks/forge/run-production/material-consumption>

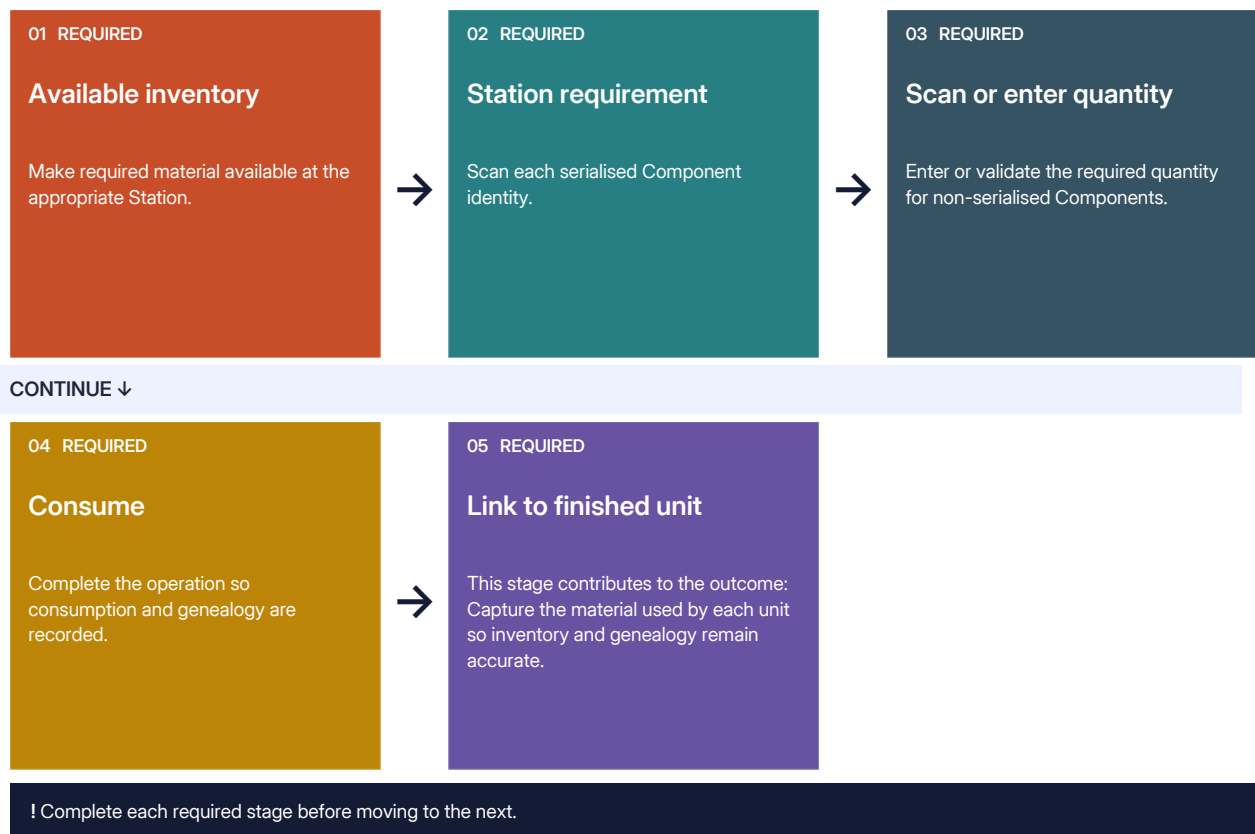
Requirement	Required
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Before you begin

- Operator Execution Flow
- Serial, Lot, and Mixed Material Tracking

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Serialised Components cannot be substituted with a quantity-only entry.

What to do

- 1 Make required material available at the appropriate Station.
- 2 Scan each serialised Component identity.
- 3 Enter or validate the required quantity for non-serialised Components.
- 4 Complete the operation so consumption and genealogy are recorded.

Rules to remember

- Serialised Components cannot be substituted with a quantity-only entry.
- Quantity consumption must match the configured requirement.
- Material should be transferred before the unit reaches the Station that consumes it.

Ready when

- ☐ Station inventory covers current production.
- ☐ Serialised Components are uniquely captured.
- ☐ Consumption is visible in material traceability.

Statuses and Route Logs

Read the history Forge creates as Work Orders and units move through production.

<https://www.fynd.com/cookbooks/forge/run-production/statuses-and-route-logs>

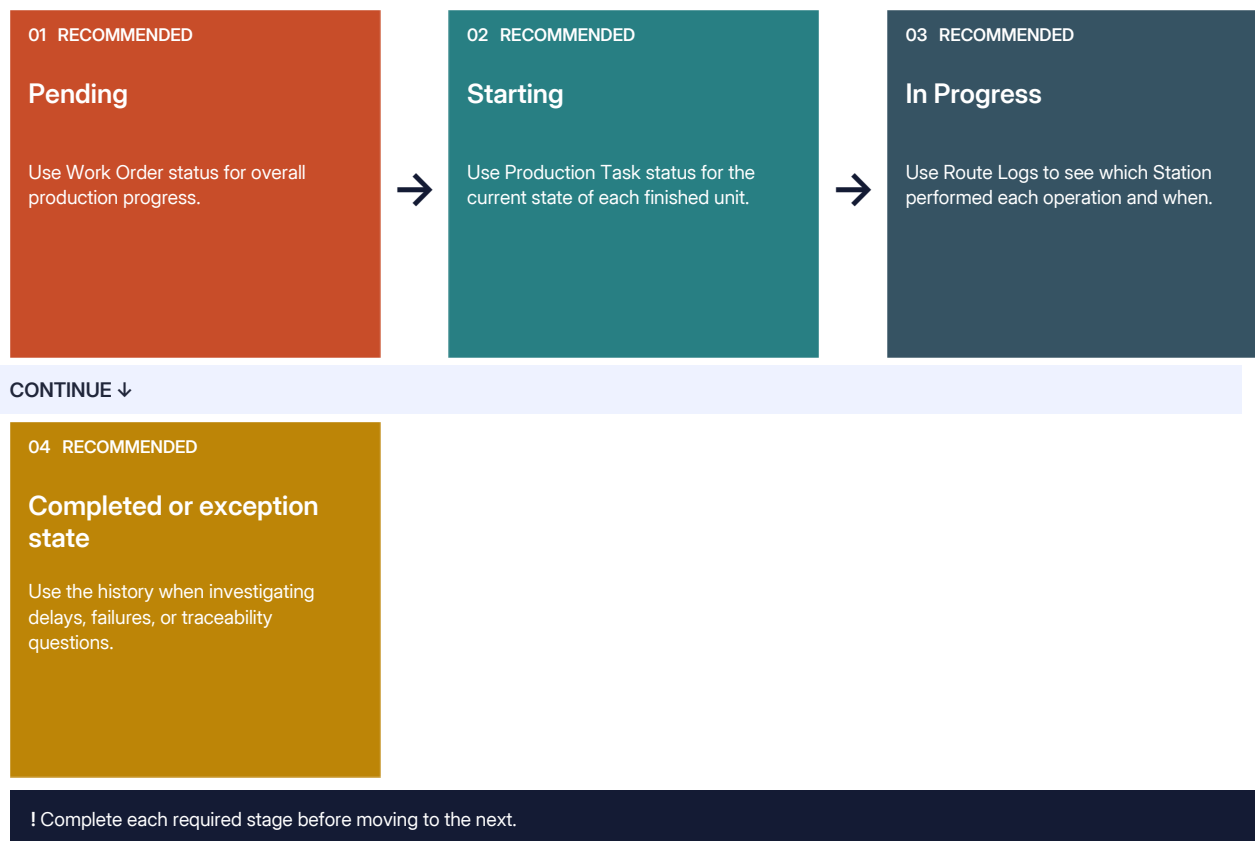
Requirement	Recommended
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Before you begin

- Production Tasks

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Work Order status and Task status answer different questions.

What to do

- 1 Use Work Order status for overall production progress.
- 2 Use Production Task status for the current state of each finished unit.
- 3 Use Route Logs to see which Station performed each operation and when.
- 4 Use the history when investigating delays, failures, or traceability questions.

Rules to remember

- Work Order status and Task status answer different questions.
- Route Logs are the operation-level evidence of what happened.
- Hold, repair, rework, and scrap can interrupt the normal progression.

Ready when

- [] Users know where to check overall and unit-level status.
- [] Completed operations appear in Route Logs.
- [] Exception states have a clear next action.

GUIDED STAGE

Quality and Exceptions

Configure inspections and handle failures through defects, NCR, repair, rework, hold, scrap, and teardown.

<https://www.fynd.com/cookbooks/forge/quality-exceptions>

Follow this stage in order

- 1 [Configure Quality Controls](#) - Decide where inspection, sampling, and verification belong in the manufacturing route.
- 2 [Inspections, Defects, NCR, and CAPA](#) - Understand the escalation path from a failed check to controlled corrective action.
- 3 [Repair and Rework](#) - Return failed units to a controlled process without losing their production history.
- 4 [Hold, Scrap, and Teardown](#) - Choose the correct action when a unit must pause, terminate, or be disassembled.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Configure Quality Controls

Decide where inspection, sampling, and verification belong in the manufacturing route.

<https://www.fynd.com/cookbooks/forge/quality-exceptions/quality-controls>

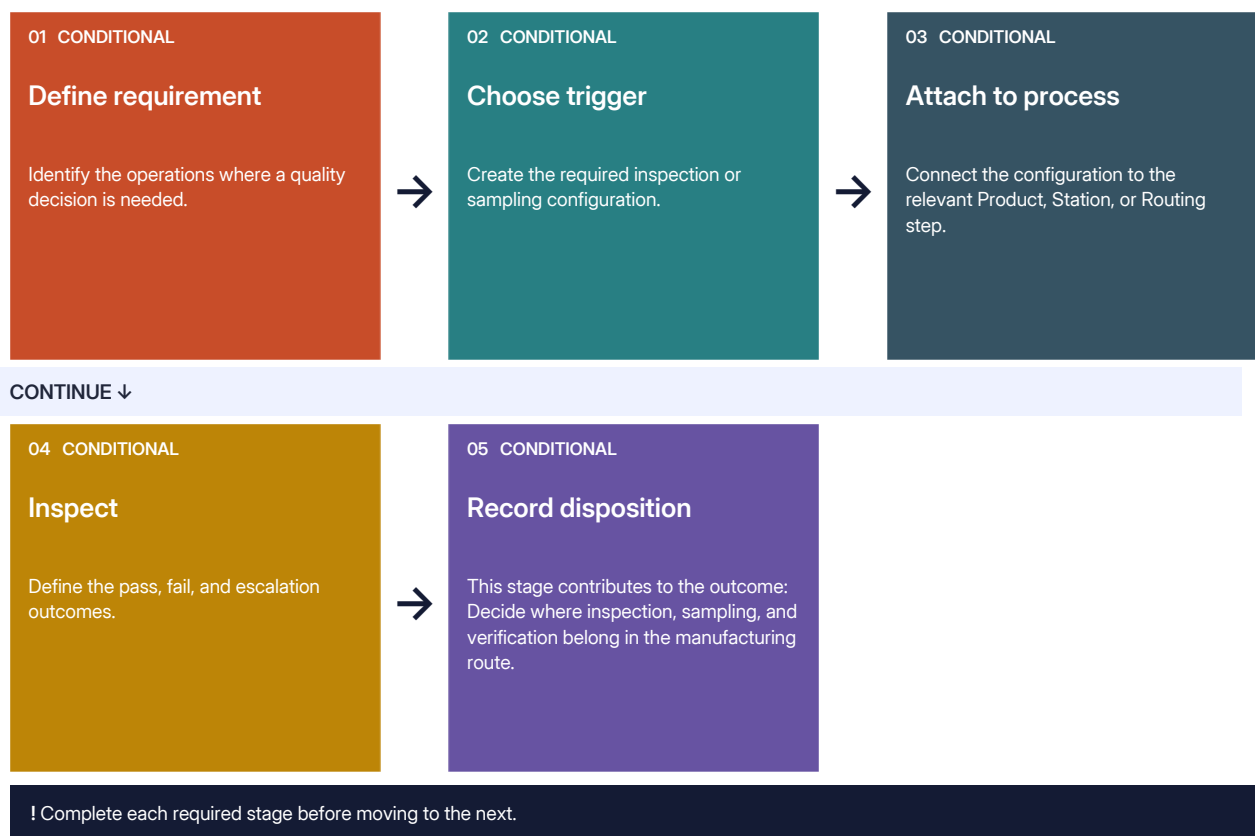
Requirement	Conditional
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Before you begin

- Create and Validate a Routing

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Quality controls should be placed where evidence can be collected reliably.

What to do

- 1 Identify the operations where a quality decision is needed.
- 2 Create the required inspection or sampling configuration.
- 3 Connect the configuration to the relevant Product, Station, or Routing step.
- 4 Define the pass, fail, and escalation outcomes.

Rules to remember

- Quality controls should be placed where evidence can be collected reliably.
- Sampling reduces inspection volume but does not remove traceability.
- A failed result needs a configured disposition path.

Ready when

- ☐ Inspection criteria are measurable.
- ☐ Triggers and sample sizes are understood.
- ☐ Failure outcomes are configured before production.

Inspections, Defects, NCR, and CAPA

Understand the escalation path from a failed check to controlled corrective action.

<https://www.fynd.com/cookbooks/forge/quality-exceptions/inspection-to-capa>

Requirement	Conditional
-------------	-------------

Before you begin

- Configure Quality Controls

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A Defect describes the observed problem; an NCR controls the nonconforming case.

What to do

- 1 Use an Inspection to record whether defined criteria passed or failed.
- 2 Create or link a Defect when a specific nonconformity is found.
- 3 Use an NCR to control and investigate nonconforming output or material.
- 4 Use a Disposition to decide what happens to the affected unit.
- 5 Use CAPA when the issue requires broader corrective or preventive action.

Rules to remember

- A Defect describes the observed problem; an NCR controls the nonconforming case.
- A Disposition is the decision, while approval controls who can authorise it.
- CAPA is for systemic correction, not every isolated failure.

Ready when

- ☐ The correct record type is used for the issue.
- ☐ Affected units and material are traceable.
- ☐ Disposition and approval are complete before release.

Repair and Rework

Return failed units to a controlled process without losing their production history.

<https://www.fynd.com/cookbooks/forge/quality-exceptions/repair-and-rework>

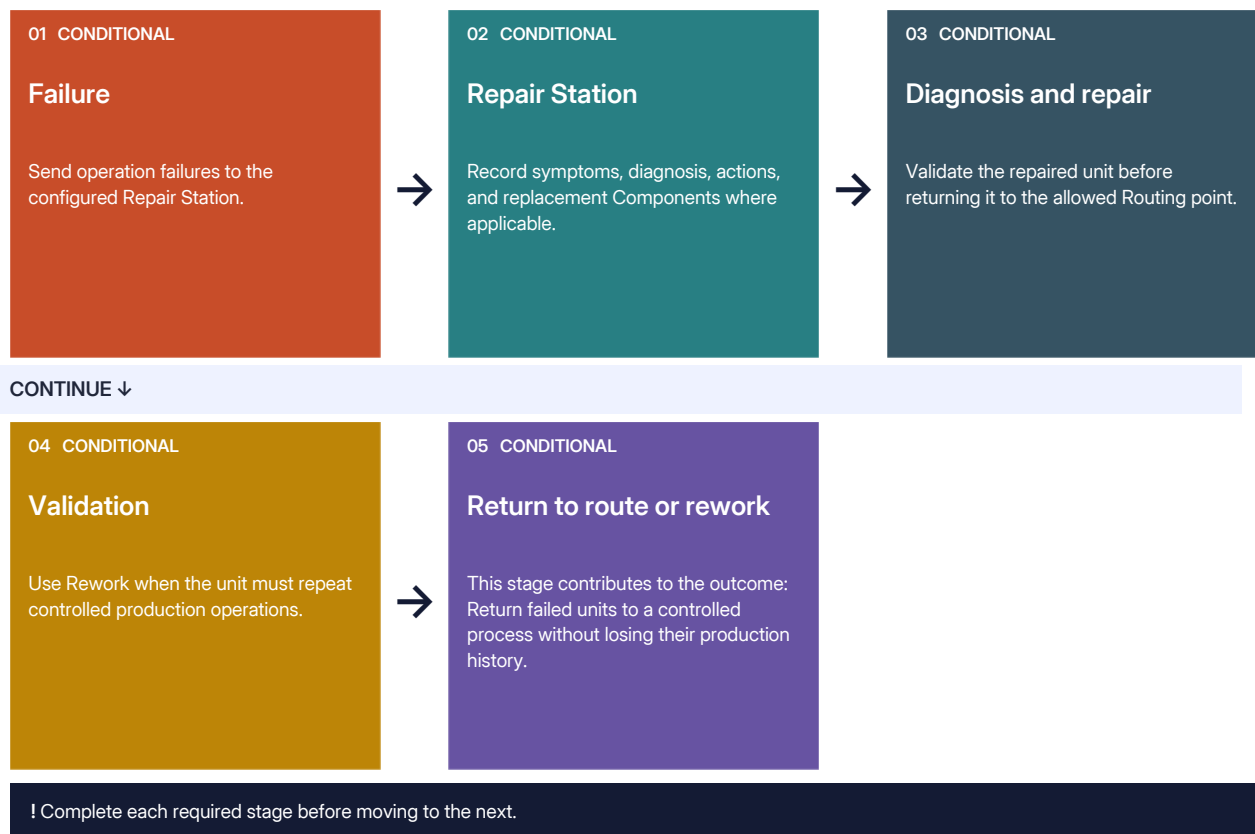
Requirement	Conditional
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Before you begin

- Create Stations and Repair Stations
- Create and Validate a Routing

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Repair fixes the failed unit; Rework repeats or redirects manufacturing work.

What to do

- 1 Send operation failures to the configured Repair Station.
- 2 Record symptoms, diagnosis, actions, and replacement Components where applicable.
- 3 Validate the repaired unit before returning it to the allowed Routing point.
- 4 Use Rework when the unit must repeat controlled production operations.

Rules to remember

- Repair fixes the failed unit; Rework repeats or redirects manufacturing work.
- The repair destination must be different from the failed operation Station.
- Return targets must be allowed by the configured route and failure logic.

Ready when

- ☐ Repair records preserve the original unit identity.
- ☐ Replacement material is traceable.
- ☐ The unit returns only to an allowed operation.

Hold, Scrap, and Teardown

Choose the correct action when a unit must pause, terminate, or be disassembled.

<https://www.fynd.com/cookbooks/forge/quality-exceptions/hold-scrap-teardown>

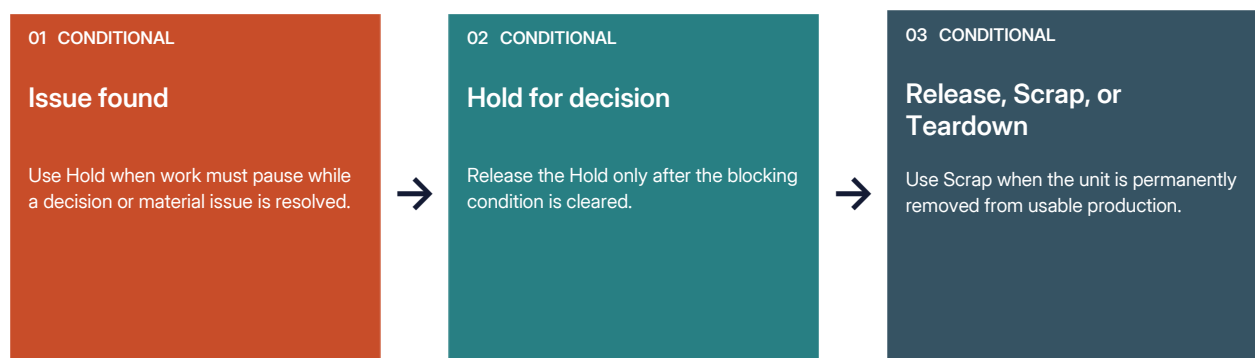
Requirement	Conditional
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Before you begin

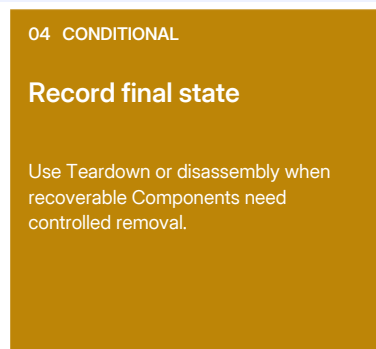
- Statuses and Route Logs

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



CONTINUE ↓



! Complete each required stage before moving to the next.

Why this connection matters: Hold is reversible; Scrap is a terminal production outcome.

What to do

- 1 Use Hold when work must pause while a decision or material issue is resolved.
- 2 Release the Hold only after the blocking condition is cleared.
- 3 Use Scrap when the unit is permanently removed from usable production.
- 4 Use Teardown or disassembly when recoverable Components need controlled removal.

Rules to remember

- Hold is reversible; Scrap is a terminal production outcome.
- Scrap requires a clear reason and preserved traceability.
- Recovered serialised Components need an explicit inventory and genealogy outcome.

Ready when

- ☐ The chosen action matches the physical outcome.
- ☐ Reason codes and notes are complete.
- ☐ Inventory and traceability reflect the final decision.

GUIDED STAGE

Package, Ship, and Trace

Configure container operations, verify shipment readiness, and trace finished units and materials across their lifecycle.

<https://www.fynd.com/cookbooks/forge/package-trace>

Follow this stage in order

- 1 [Packaging in Forge](#) - Understand how unit, container, label, weight, and closure operations fit into a Routing.
- 2 [Containers and Labels](#) - Create a clear packaging identity from individual units through cartons and pallets.
- 3 [Shipment Verification](#) - Verify that packaged output matches shipment requirements before dispatch.
- 4 [Traceability, Genealogy, and Recall](#) - Follow a finished unit back to its Work Order, operations, Components, material lots, containers, and shipment.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Packaging in Forge

Understand how unit, container, label, weight, and closure operations fit into a Routing.

<https://www.fynd.com/cookbooks/forge/package-trace/packaging>

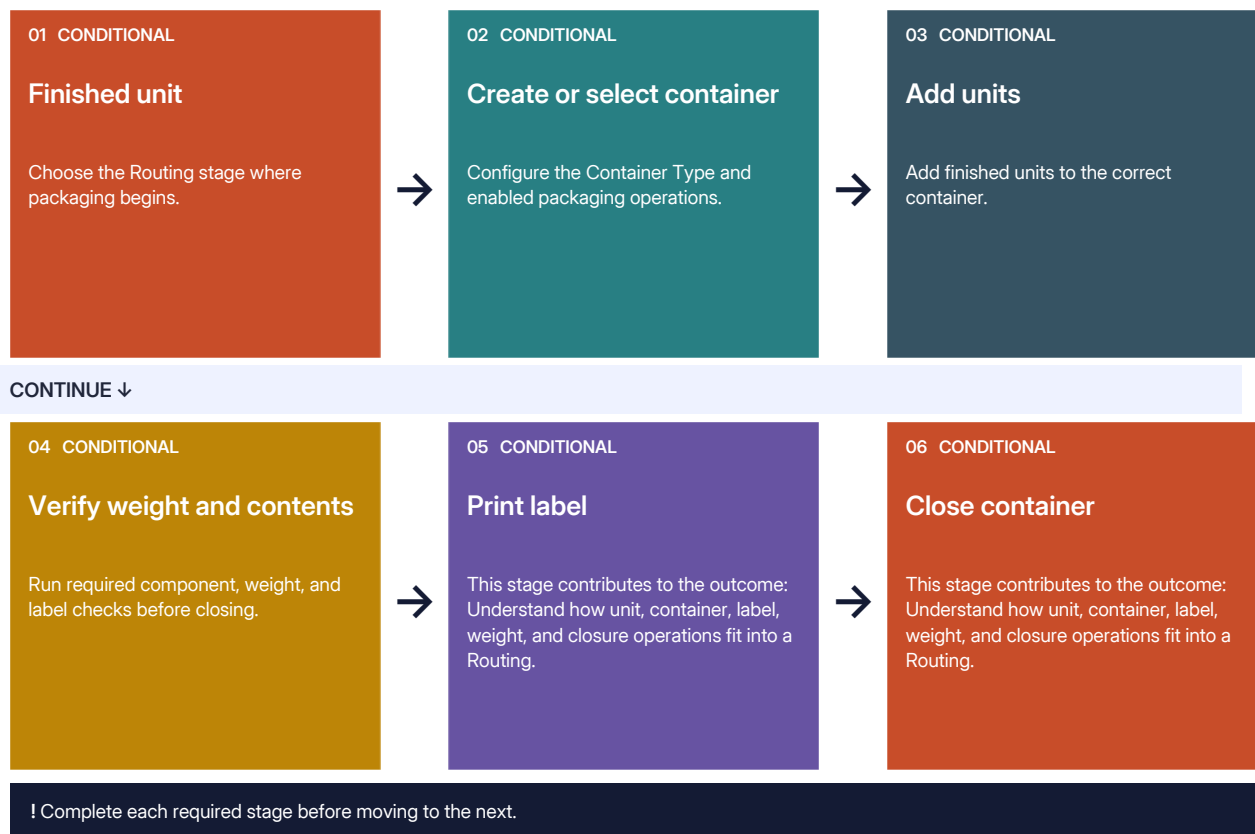
Requirement	Conditional
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Before you begin

- Create and Validate a Routing

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Packaging operations are required only when the Product uses the configured container flow.

What to do

- 1 Choose the Routing stage where packaging begins.
- 2 Configure the Container Type and enabled packaging operations.
- 3 Add finished units to the correct container.
- 4 Run required component, weight, and label checks before closing.

Rules to remember

- Packaging operations are required only when the Product uses the configured container flow.
- At least one packaging operation must be enabled when stage packaging is configured.
- Auto-close depends on label printing; auto-print after weight depends on weight validation.

Ready when

- ☐ Container hierarchy matches the physical packaging process.
- ☐ Required labels and printers are ready.
- ☐ Weight and closure rules are testable.

Containers and Labels

Create a clear packaging identity from individual units through cartons and pallets.

<https://www.fynd.com/cookbooks/forge/package-trace/containers-and-labels>

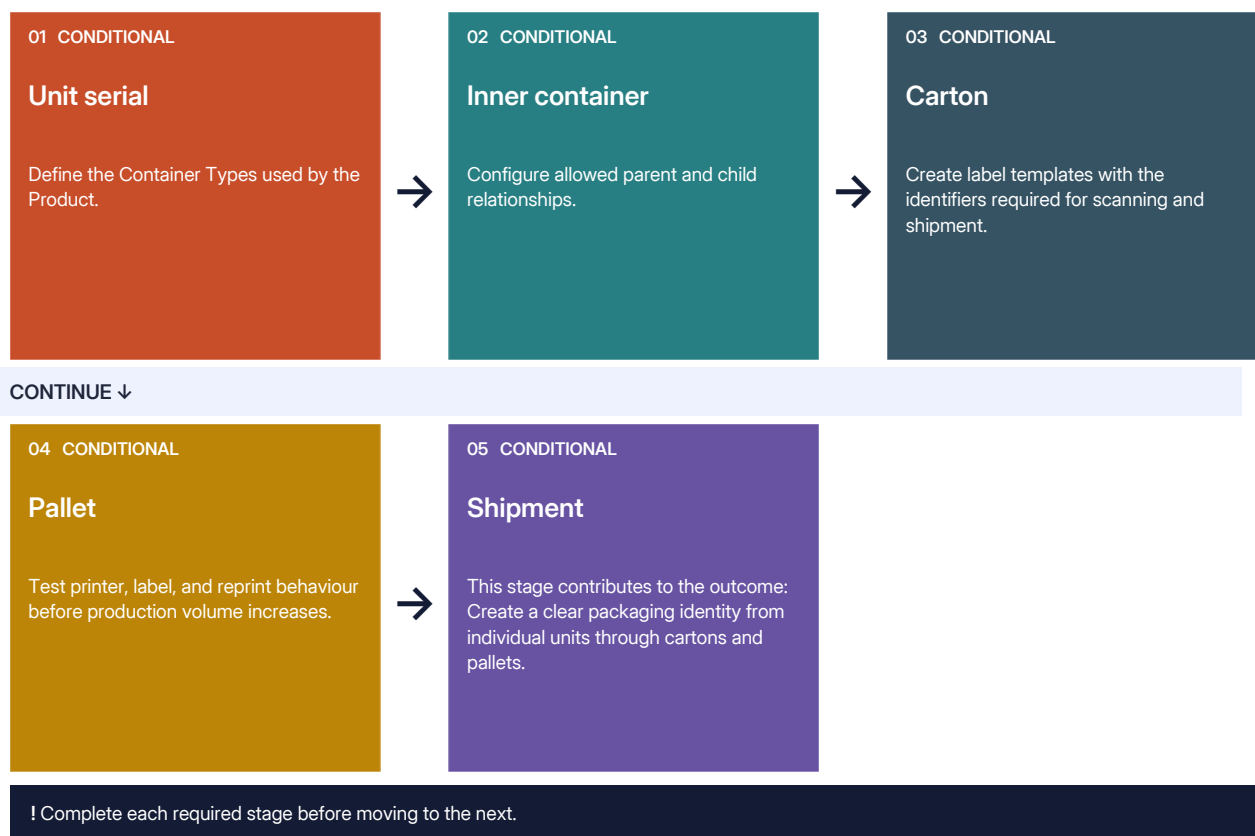
Requirement	Conditional
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Before you begin

- Packaging in Forge

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A container identity must remain unique and scannable.

What to do

- 1 Define the Container Types used by the Product.
- 2 Configure allowed parent and child relationships.
- 3 Create label templates with the identifiers required for scanning and shipment.
- 4 Test printer, label, and reprint behaviour before production volume increases.

Rules to remember

- A container identity must remain unique and scannable.
- Closed or abandoned containers should not accept new units.
- Reprints should preserve auditability and follow configured approval rules.

Ready when

- ☐ Container levels mirror physical packaging.
- ☐ Labels include the required identifiers.
- ☐ Scan and reprint paths have been tested.

Shipment Verification

Verify that packaged output matches shipment requirements before dispatch.

<https://www.fynd.com/cookbooks/forge/package-trace/shipment-verification>

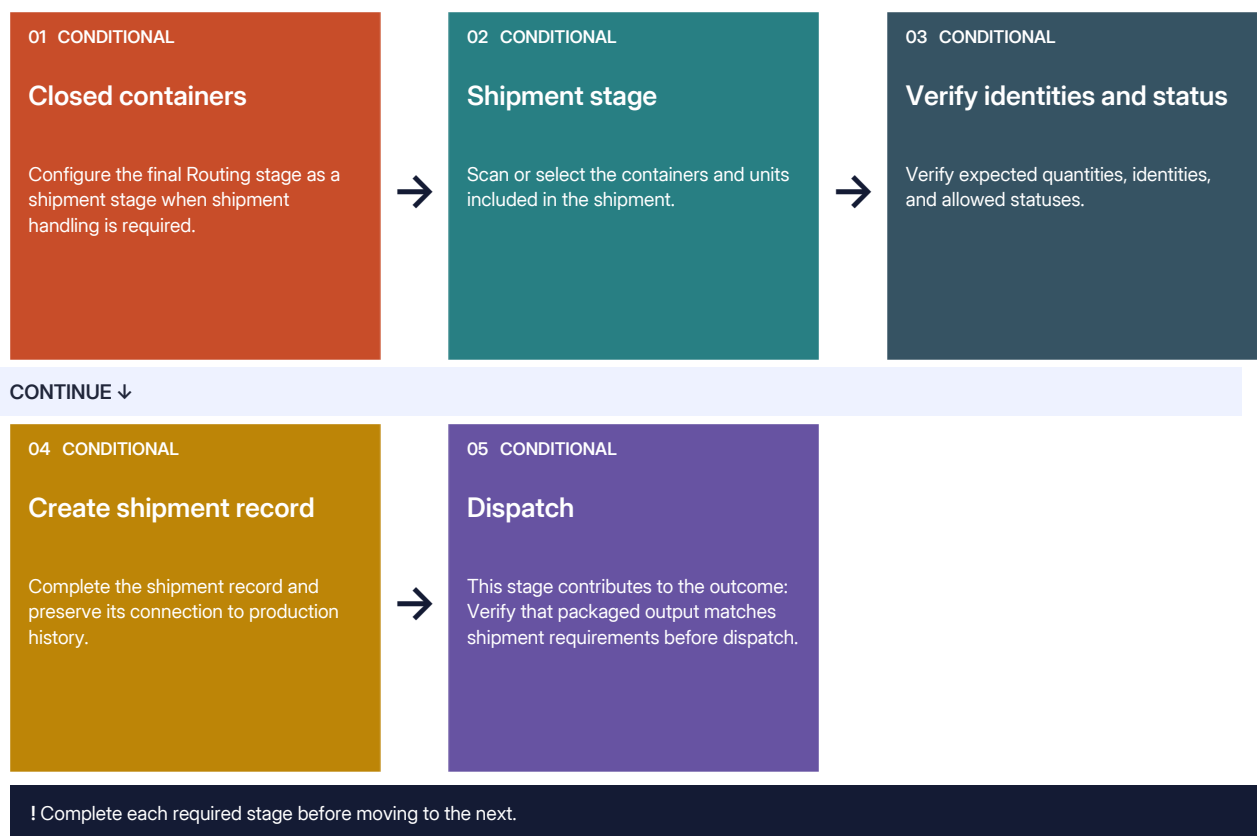
Requirement	Conditional
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Before you begin

- Containers and Labels

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Shipment verification should use closed, valid containers.

What to do

- 1 Configure the final Routing stage as a shipment stage when shipment handling is required.
- 2 Scan or select the containers and units included in the shipment.
- 3 Verify expected quantities, identities, and allowed statuses.
- 4 Complete the shipment record and preserve its connection to production history.

Rules to remember

- Shipment verification should use closed, valid containers.
- Abandoned packaging identities must not satisfy verification gates.
- Shipment records should remain linked to the Product and unit genealogy.

Ready when

- ☐ Expected and scanned shipment quantities match.
- ☐ All included containers have valid status.
- ☐ The shipment can be traced back to Work Orders and units.

Traceability, Genealogy, and Recall

Follow a finished unit back to its Work Order, operations, Components, material lots, containers, and shipment.

<https://www.fynd.com/cookbooks/forge/package-trace/traceability-genealogy-recall>

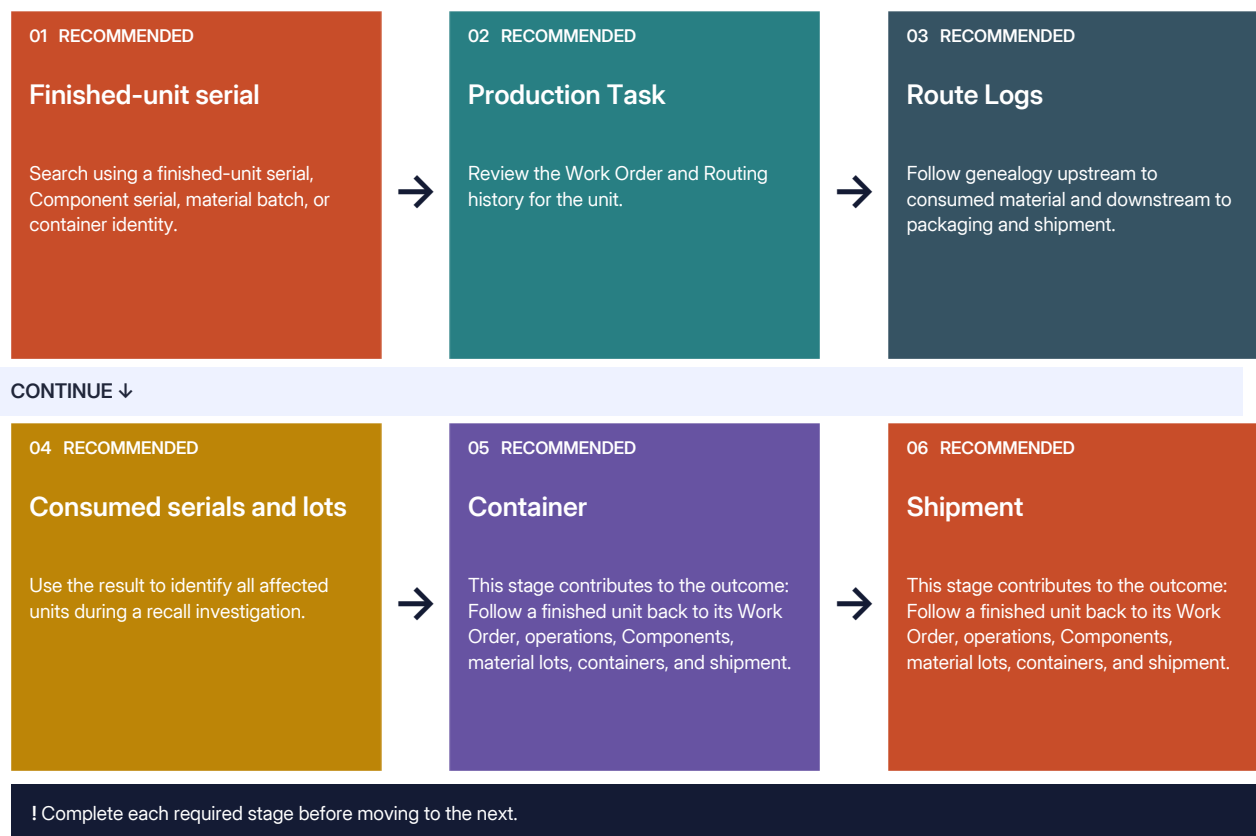
Requirement	Recommended
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Before you begin

- Material Consumption During Production
- Shipment Verification

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Traceability quality depends on accurate scanning and consumption during production.

What to do

- 1 Search using a finished-unit serial, Component serial, material batch, or container identity.
- 2 Review the Work Order and Routing history for the unit.

- 3 Follow genealogy upstream to consumed material and downstream to packaging and shipment.
- 4 Use the result to identify all affected units during a recall investigation.

Rules to remember

- Traceability quality depends on accurate scanning and consumption during production.
- Lot genealogy and unit genealogy answer different but connected questions.
- A recall notice controls the response; traceability identifies the affected scope.

Ready when

- ☐ A trial unit can be traced from input material to shipment.
- ☐ Material batches identify all affected finished units.
- ☐ Recall searches return a clear, reviewable scope.

GUIDED STAGE

Operate and Improve

Use dashboards, shift handovers, maintenance, tools, reports, and audit history to keep production controlled.

<https://www.fynd.com/cookbooks/forge/operate-improve>

Follow this stage in order

- 1 [Dashboards and Production Monitoring](#) - Choose the dashboard that answers the operational question instead of treating every metric as the same.
- 2 [Shift Operations and Handovers](#) - Maintain continuity when production responsibility moves from one Shift to another.
- 3 [Tools, Equipment, and Maintenance](#) - Keep required production assets available, correctly located, and safe to use.
- 4 [Reports, Exports, and Audit Trail](#) - Use reports and immutable history to explain production results and configuration changes.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Dashboards and Production Monitoring

Choose the dashboard that answers the operational question instead of treating every metric as the same.

<https://www.fynd.com/cookbooks/forge/operate-improve/dashboards>

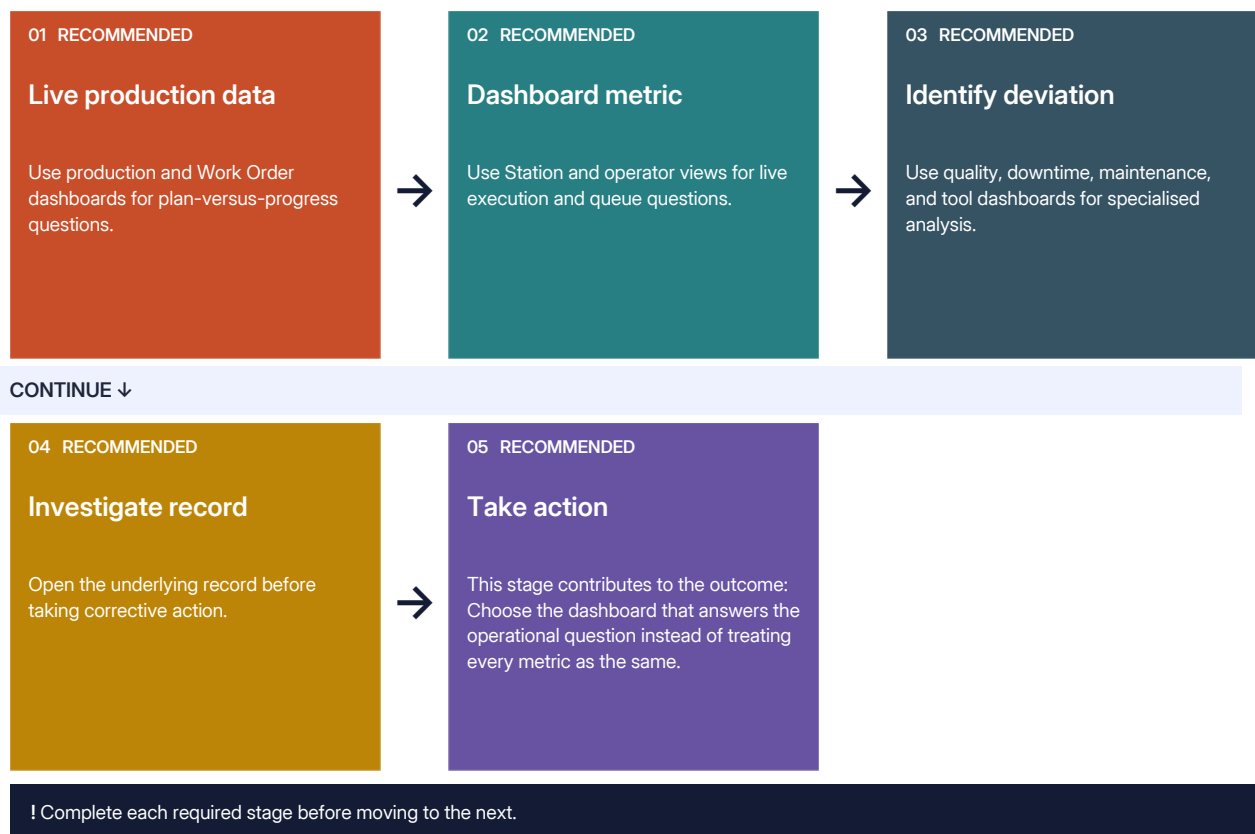
Requirement	Recommended
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Before you begin

- Statuses and Route Logs

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: A dashboard summarises records; the underlying records remain the source of evidence.

What to do

- 1 Use production and Work Order dashboards for plan-versus-progress questions.
- 2 Use Station and operator views for live execution and queue questions.
- 3 Use quality, downtime, maintenance, and tool dashboards for specialised analysis.

- 4 Open the underlying record before taking corrective action.

Rules to remember

- A dashboard summarises records; the underlying records remain the source of evidence.
- Use consistent time, Plant, Line, Product, and Shift filters when comparing metrics.
- Do not compare metrics with different definitions or time windows.

Ready when

- ☐ Each operational meeting uses agreed dashboards.
- ☐ Metric definitions are understood.
- ☐ Users can move from a dashboard exception to the underlying record.

Shift Operations and Handovers

Maintain continuity when production responsibility moves from one Shift to another.

<https://www.fynd.com/cookbooks/forge/operate-improve/shift-operations>

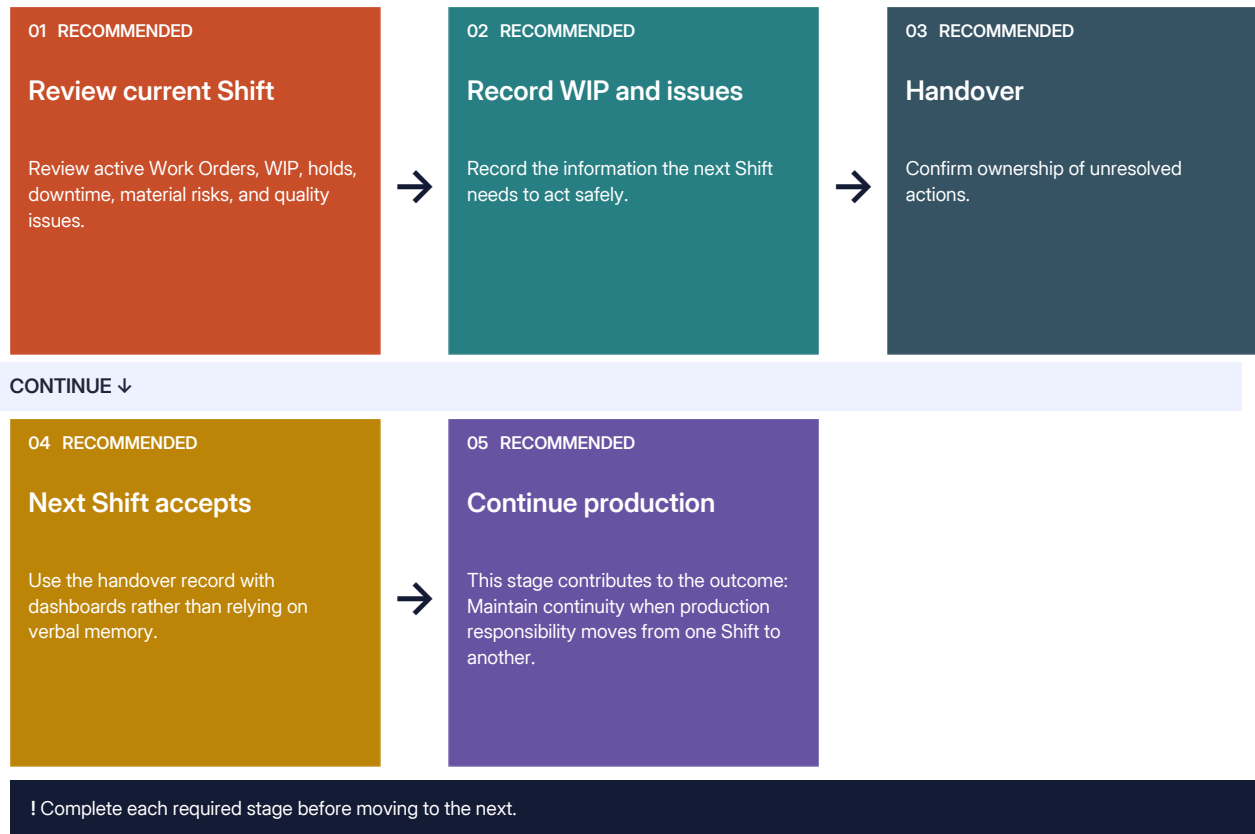
Requirement	Recommended
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Before you begin

- Create Shifts
- Dashboards and Production Monitoring

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Handover information should be specific to records and Lines.

What to do

- 1 Review active Work Orders, WIP, holds, downtime, material risks, and quality issues.
- 2 Record the information the next Shift needs to act safely.
- 3 Confirm ownership of unresolved actions.
- 4 Use the handover record with dashboards rather than relying on verbal memory.

Rules to remember

- Handover information should be specific to records and Lines.
- Open quality, maintenance, and material blockers must be visible.
- A new Shift should not restart work that is already in progress.

Ready when

- ☐ WIP and blocked units are identified.
- ☐ Open actions have clear next steps.
- ☐ The next Shift can continue without reconstructing the previous Shift.

Tools, Equipment, and Maintenance

Keep required production assets available, correctly located, and safe to use.

<https://www.fynd.com/cookbooks/forge/operate-improve/tools-and-maintenance>

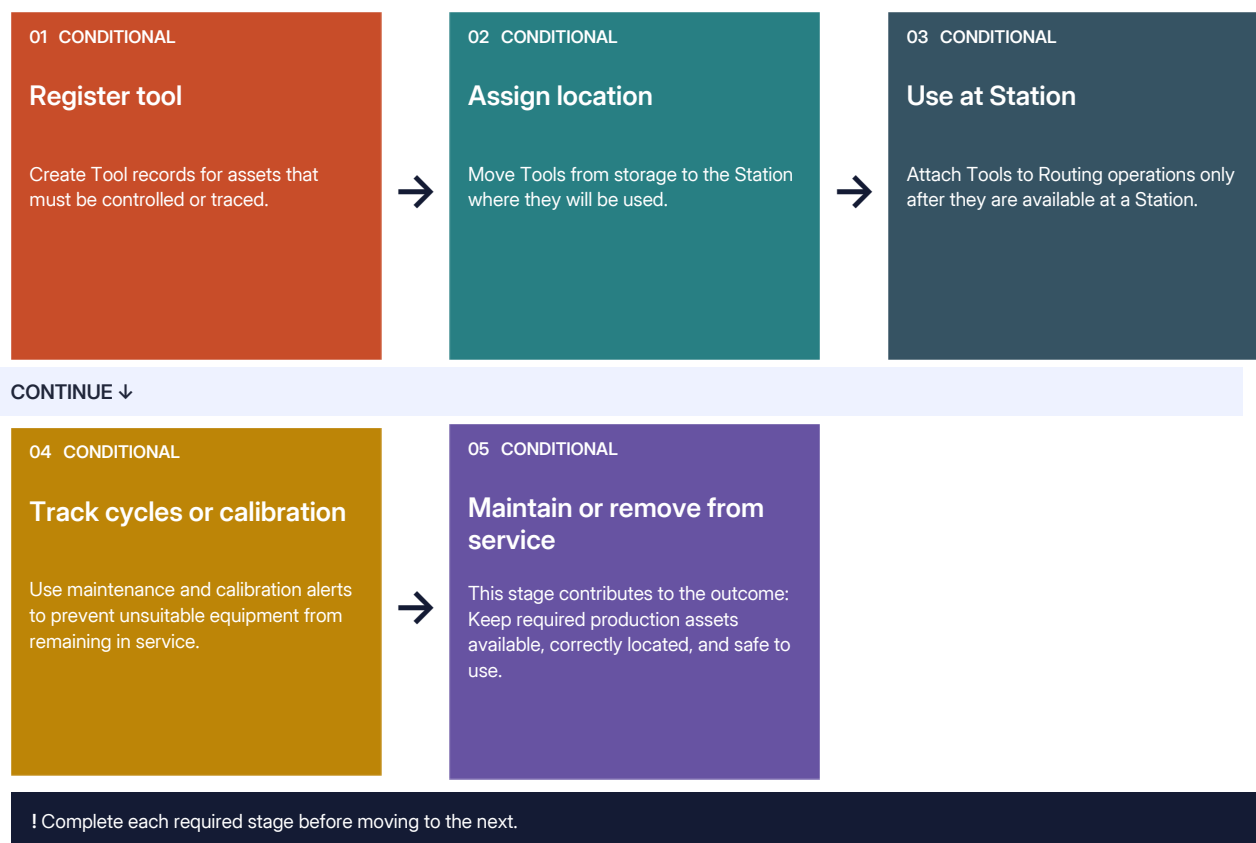
Requirement	Conditional
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Before you begin

- Create Stations and Repair Stations

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Tools in storage cannot be added to a Routing operation.

What to do

- 1 Create Tool records for assets that must be controlled or traced.
- 2 Move Tools from storage to the Station where they will be used.
- 3 Attach Tools to Routing operations only after they are available at a Station.
- 4 Use maintenance and calibration alerts to prevent unsuitable equipment from remaining in service.

Rules to remember

- Tools in storage cannot be added to a Routing operation.
- Tool use should be recorded where cycle, calibration, or traceability matters.
- Maintenance state should prevent unsafe or invalid production use.

Ready when

- ☐ Required Tools are registered and located correctly.
- ☐ Calibration and maintenance dates are current.
- ☐ Routing requirements match available assets.

Reports, Exports, and Audit Trail

Use reports and immutable history to explain production results and configuration changes.

<https://www.fynd.com/cookbooks/forge/operate-improve/reports-and-audit>

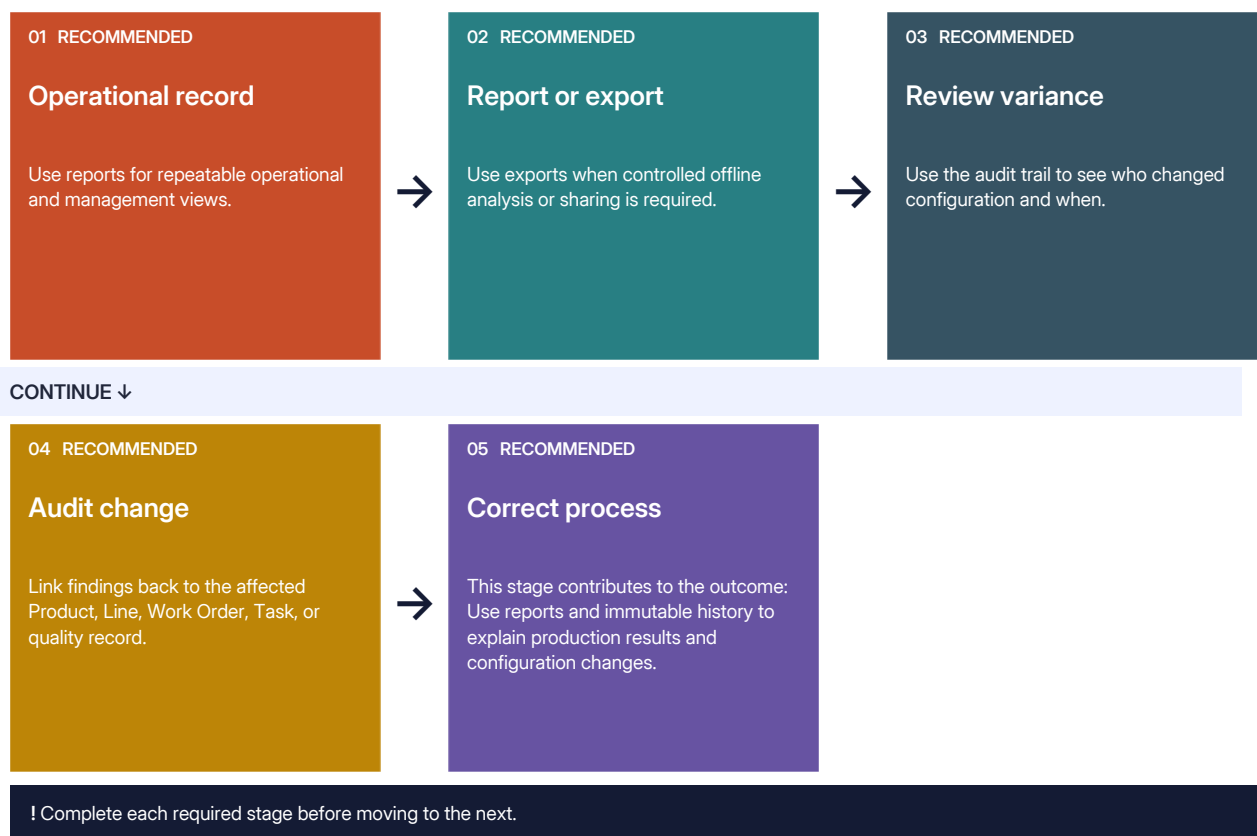
Requirement	Recommended
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Before you begin

- Dashboards and Production Monitoring

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Exports are snapshots and can become outdated.

What to do

- 1 Use reports for repeatable operational and management views.
- 2 Use exports when controlled offline analysis or sharing is required.
- 3 Use the audit trail to see who changed configuration and when.
- 4 Link findings back to the affected Product, Line, Work Order, Task, or quality record.

Rules to remember

- Exports are snapshots and can become outdated.
- Audit history explains a change but does not replace configuration review.
- Sensitive or internal fields should not be added to the public cookbook.

Ready when

- ☐ Key reports have agreed filters and definitions.
- ☐ Exports are handled according to data policy.
- ☐ Configuration changes can be reconstructed from audit history.

GUIDED STAGE

Help and Reference

Use checklists, requirement matrices, terminology, and decision trees when setting up or troubleshooting Forge.

<https://www.fynd.com/cookbooks/forge/help-reference>

Follow this stage in order

- 1 [Required, Recommended, and Conditional Setup](#) - Use one matrix to understand which records block production and which improve control.
- 2 [First Production Run Checklist](#) - Run a small controlled Work Order to validate the complete configuration before scaling volume.
- 3 [Troubleshooting Decision Tree](#) - Start with the failing action and work backward through its dependencies.
- 4 [Forge MES Glossary](#) - Use globally neutral definitions for the core records and actions in Forge MES.
- 5 [Capability Availability Guide](#) - Recognise which capabilities are always part of the core flow and which appear only when configured.

Why the order matters

Each topic prepares information used by the next one. Complete the required topics and their readiness checks before moving forward.

Required, Recommended, and Conditional Setup

Use one matrix to understand which records block production and which improve control.

<https://www.fynd.com/cookbooks/forge/help-reference/requirement-matrix>

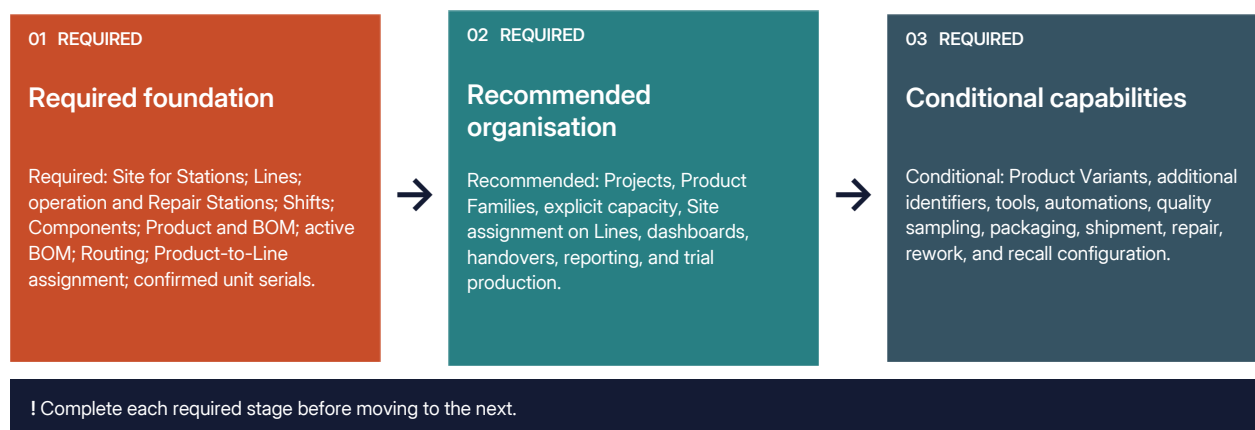
Requirement	Required
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Before you begin

- Complete Forge Setup Journey

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Required labels reflect the supported normal production journey.

What to do

- 1 Required: Site for Stations; Lines; operation and Repair Stations; Shifts; Components; Product and BOM; active BOM; Routing; Product-to-Line assignment; confirmed unit serials.
- 2 Recommended: Projects, Product Families, explicit capacity, Site assignment on Lines, dashboards, handovers, reporting, and trial production.
- 3 Conditional: Product Variants, additional identifiers, tools, automations, quality sampling, packaging, shipment, repair, rework, and recall configuration.

Rules to remember

- Required labels reflect the supported normal production journey.
- Recommended records may not be database blockers but prevent avoidable operational confusion.
- Conditional features become required when the selected production design depends on them.

Ready when

- ☐ Required setup is complete.
- ☐ Recommended structure has been reviewed.
- ☐ Every conditional capability has an explicit use or exclusion decision.

First Production Run Checklist

Run a small controlled Work Order to validate the complete configuration before scaling volume.

<https://www.fynd.com/cookbooks/forge/help-reference/first-production-run>

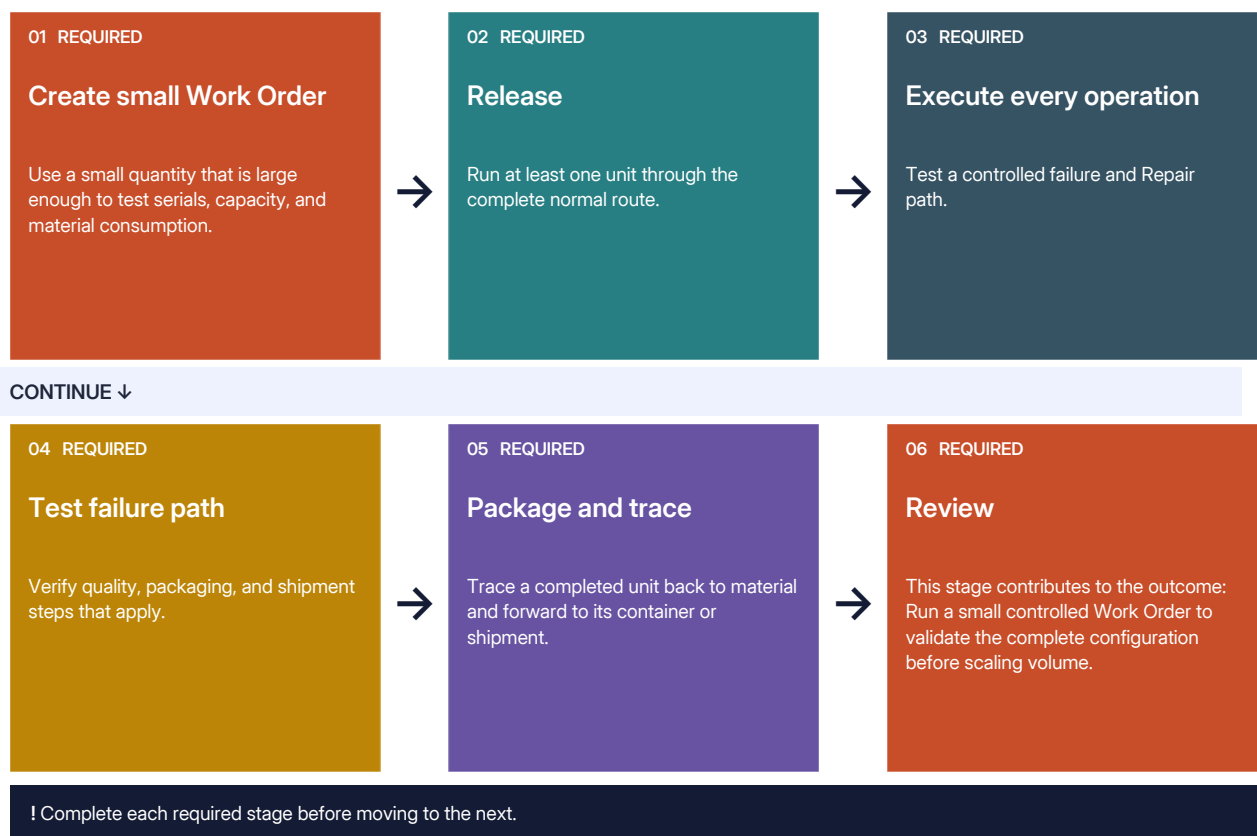
Requirement	Required
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Before you begin

- Pre-Start Validation and Release

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Do not use the first full-volume order as the first complete process test.

What to do

- 1 Use a small quantity that is large enough to test serials, capacity, and material consumption.
- 2 Run at least one unit through the complete normal route.
- 3 Test a controlled failure and Repair path.
- 4 Verify quality, packaging, and shipment steps that apply.
- 5 Trace a completed unit back to material and forward to its container or shipment.

Rules to remember

- Do not use the first full-volume order as the first complete process test.
- Record every configuration issue and retest after correction.
- The trial is complete only when the unit history is understandable end to end.

Ready when

- ☐ Work Order creates the expected Production Tasks.
- ☐ Normal and failure paths work.
- ☐ Material consumption and inventory are correct.
- ☐ Packaging and shipment work where configured.
- ☐ End-to-end traceability is complete.

Troubleshooting Decision Tree

Start with the failing action and work backward through its dependencies.

<https://www.fynd.com/cookbooks/forgem/helppreference/troubleshooting>

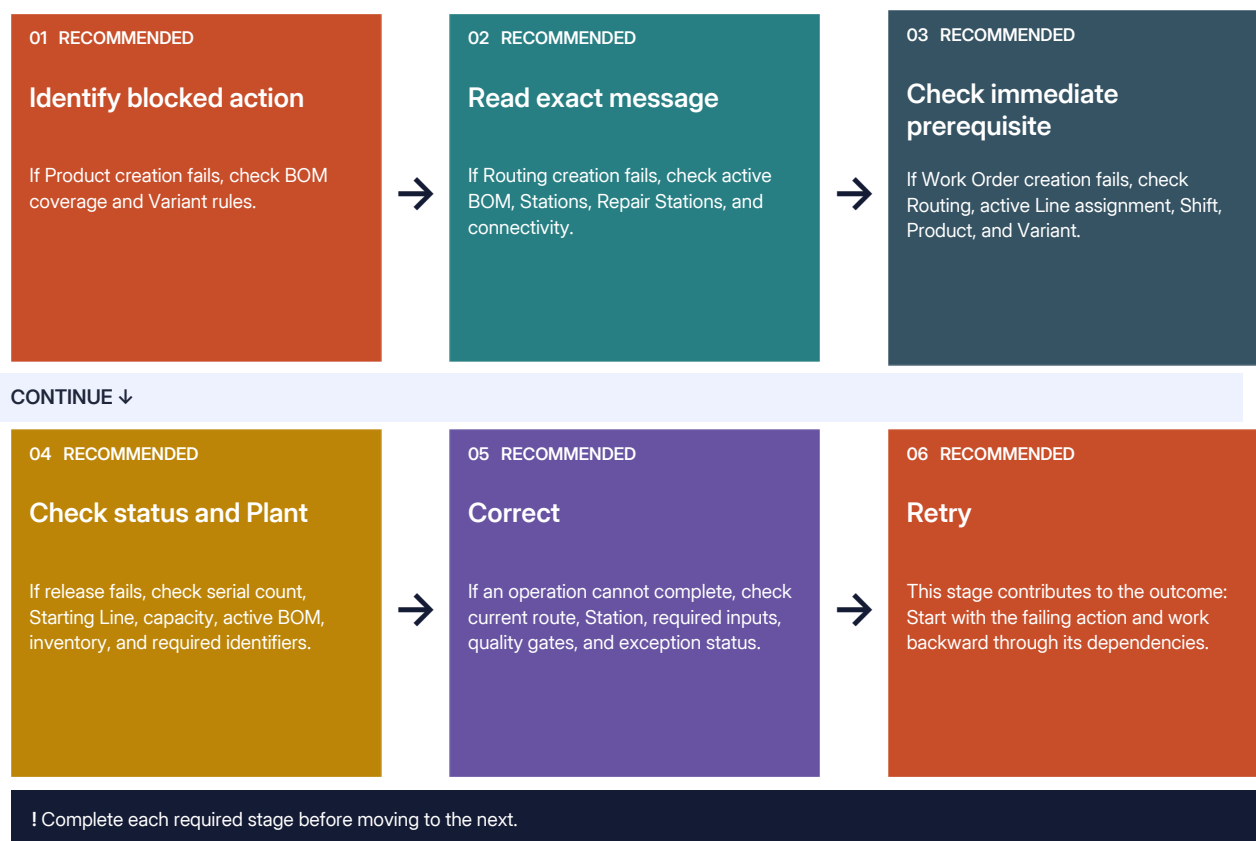
Requirement	Recommended
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Before you begin

- How Forge Data Connects

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Keep the exact error message; do not troubleshoot from memory.

What to do

- 1 If Product creation fails, check BOM coverage and Variant rules.
- 2 If Routing creation fails, check active BOM, Stations, Repair Stations, and connectivity.
- 3 If Work Order creation fails, check Routing, active Line assignment, Shift, Product, and Variant.
- 4 If release fails, check serial count, Starting Line, capacity, active BOM, inventory, and required identifiers.

- 5 If an operation cannot complete, check current route, Station, required inputs, quality gates, and exception status.

Rules to remember

- Keep the exact error message; do not troubleshoot from memory.
- Confirm the record belongs to the current Plant.
- Fix the earliest missing dependency instead of changing downstream data.

Ready when

- [] The blocked action is clearly identified.
- [] The immediate prerequisite has been checked.
- [] The correction is verified with the smallest safe retry.

Forge MES Glossary

Use globally neutral definitions for the core records and actions in Forge MES.

<https://www.fynd.com/cookbooks/forge/help-reference/glossary>

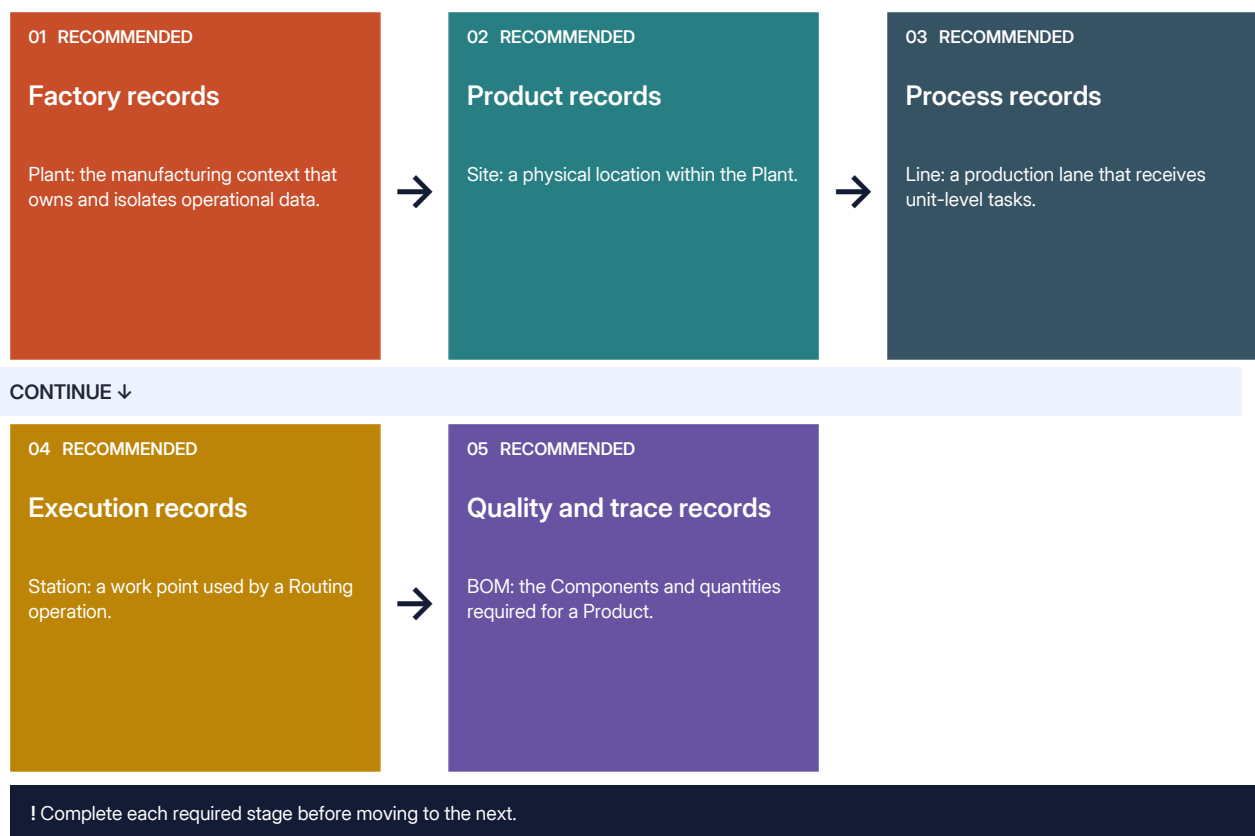
Requirement	Recommended
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Before you begin

- No earlier cookbook topic is required.

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Use Product for finished output and Component for input material.

What to do

- 1 Plant: the manufacturing context that owns and isolates operational data.
- 2 Site: a physical location within the Plant.
- 3 Line: a production lane that receives unit-level tasks.
- 4 Station: a work point used by a Routing operation.
- 5 BOM: the Components and quantities required for a Product.

- 6 Routing: the connected sequence of operations a unit follows.
- 7 Work Order: the executable instruction to produce a Product quantity.
- 8 Production Task: one finished unit moving through production.
- 9 Genealogy: the relationship between finished units, consumed material, packaging, and shipment.

Rules to remember

- Use Product for finished output and Component for input material.
- Use Production Order for grouping and Work Order for execution.
- Use Repair for fixing a failed unit and Rework for repeating controlled manufacturing operations.

Ready when

- [] Terminology is used consistently in procedures and training.
- [] Local factory terms are mapped to the Forge term.
- [] Users can distinguish similarly named records.

Capability Availability Guide

Recognise which capabilities are always part of the core flow and which appear only when configured.

<https://www.fynd.com/cookbooks/forge/help-reference/capability-availability>

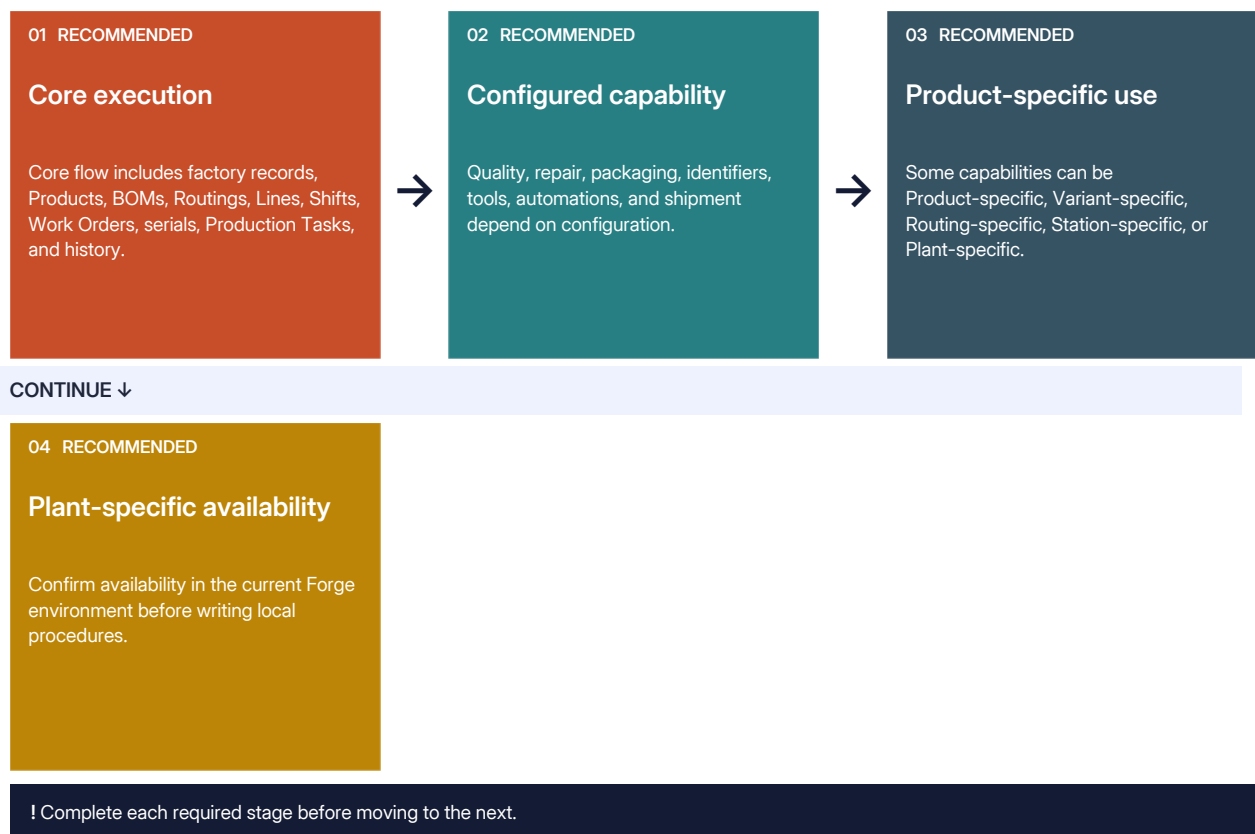
Requirement	Recommended
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Before you begin

- Forge MES in Plain Language

How it works

Follow the arrows from left to right. Each stage prepares the information the next stage needs.



Why this connection matters: Do not assume every menu item is required for every manufacturing process.

What to do

- 1 Core flow includes factory records, Products, BOMs, Routings, Lines, Shifts, Work Orders, serials, Production Tasks, and history.
- 2 Quality, repair, packaging, identifiers, tools, automations, and shipment depend on configuration.
- 3 Some capabilities can be Product-specific, Variant-specific, Routing-specific, Station-specific, or Plant-specific.

- 4 Confirm availability in the current Forge environment before writing local procedures.

Rules to remember

- Do not assume every menu item is required for every manufacturing process.
- Do not describe proposed or unverified capabilities as available.
- Use the latest verified implementation and active configuration as the source of truth.

Ready when

- ☐ Core and conditional capabilities are clearly separated.
- ☐ Configured features have been tested in the current environment.
- ☐ Local training includes only available capabilities.